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Multi-objective Optimisation of Suction Cup and Parallel Jaw Grippers

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Abstract

Keywords: Pymoo, FastAPI

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Nomenclature

α	Bayesian acquisition function (see also α in multi-suction optimization)
α	Bayesian acquisition function (see also α in two-finger optimization)
α_i	Angular position of the i-th suction cup on the circle
$\alpha_{\mathrm{EI}}(\mathbf{x})$	Expected improvement acquisition
$\boldsymbol{\theta}$	Algorithm hyperparameters (see also $\boldsymbol{\theta}$ for hyperparameter optimization)
$\boldsymbol{\theta}$	Evolutionary algorithm hyperparameters (see also $\boldsymbol{\theta}$ for hyperparameter optimization)
Δp	Pressure difference
$\Delta\mathbf{R}$	Relative rotation between two grasp orientations
δ_{intra}	Spatial threshold for selecting additional local representatives (see also clustering procedure)
$\kappa(\mathbf{x}, \mathbf{x}')$	Covariance/kernel function
Λ	Set of reference directions for decomposition-based algorithms (see also Λ in multi-suction optimization)

Λ Set of reference directions for decomposition-based algorithms (see also Λ in two-finger optimization)

λ Lebesgue measure

$\mathbb{E}[\,\cdot\,]$ Expectation operator

$\mathbf{c}_i$ Cartesian contact position of suction cup i, with $i \in \{1,2,3\}$ (see also $\mathbf{c}_i$ for suction cup positions)

$\mathbf{c}_i$ Cartesian position of the i-th suction cup in the gripper frame

$\mathbf{e}$ Axis-aligned bounding-box side lengths of the object (see also $\mathbf{e}$ in theoretical background)

$\mathbf{e}$ Bounding-box side lengths

$\mathbf{f}(\mathbf{x})$ Objective function vector

$\mathbf{f}_{\mathrm{BO}}(\mathbf{q})$ Objective observations supplied to Ax (see also $\mathbf{f}_{\mathrm{BO}}$ in two-finger optimization)

$\mathbf{f}_{\mathrm{BO}}(\mathbf{q})$ Objective observations supplied to Ax (see also $\mathbf{f}_{\mathrm{BO}}$ in two-finger optimization)

$\mathbf{f}_{\mathrm{BO}}(\mathbf{x})$ Objective observations supplied to Ax (see also $\mathbf{f}_{\mathrm{BO}}$ in multi-suction optimization)

$\mathbf{f}_{\mathrm{BO}}(\mathbf{x})$ Objective observations supplied to Ax (see also $\mathbf{f}_{\mathrm{BO}}$ in multi-suction optimization)

$\mathbf{f}_{\mathrm{EA}}(\mathbf{q})$ Minimization-form objective vector supplied to the evolutionary algorithms (see also $\mathbf{f}_{\mathrm{EA}}$ in two-finger optimization)

$\mathbf{f}_{\mathrm{EA}}(\mathbf{q})$ Minimization-form objective vector supplied to the evolutionary algorithms (see also $\mathbf{f}_{\mathrm{EA}}$ in two-finger optimization)

$\mathbf{f}_{\mathrm{EA}}(\mathbf{x})$ Minimization-form objective vector supplied to the evolutionary algorithms (see also $\mathbf{f}_{\mathrm{EA}}$ in multi-suction optimization)

$\mathbf{F}_{\mathrm{EA}}^{(g)}$ Evolutionary objective values in generation g (see also $\mathbf{F}_{\mathrm{EA}}^{(g)}$ for evolutionary objective values)

$\mathbf{g}(\mathbf{q})$ Natural multi-objective function vector containing triangle area and grasp diversity

$\mathbf{k}(\mathbf{x})$ Natural grasp-quality objective vector (see also $\mathbf{g}(\mathbf{q})$ in multi-suction optimization for similar concept)

$\mathbf{k}(\mathbf{x})$ Natural maximization objective vector (see also $\mathbf{k}(\mathbf{x})$ for grasp-quality objectives)

$\mathbf{p}$ Cartesian gripper configuration defined by the three suction cup contact positions (see also $\mathbf{p}$ for Cartesian suction cup configuration)

$\mathbf{p}$ Cartesian suction cup configuration derived from the reduced parameterization (see also $\mathbf{x}$ for generic decision variables)

$\mathbf{P}_i$ Spline interpolation point

$\mathbf{q}$ Decision vector representing a candidate gripper configuration

$\mathbf{q}$ Reduced decision variable vector defining the suction cup layout (see also $\mathbf{x}$ for generic decision variables)

$\mathbf{q}$ Relative voxel-grid displacement (see also $\mathbf{q}$ for decision variables in multi-suction optimization)

$\mathbf{r}$ Fixed hypervolume reference point (see also $\mathbf{r}$ for reference point in theoretical background)

$\mathbf{r}$ Reference point for hypervolume computation (see also $\mathbf{r}$ for reference point in theoretical background)

$\mathbf{r}$ Reference point

$\mathbf{R}_i$ Rotation matrix of the i-th grasp orientation

$\mathbf{x}$ Decision variable vector

$\mathbf{x}$ Decision vector of the two-finger gripper (see also $\mathbf{x}$ for generic decision variables)

$\mathbf{x}^{(a)} \prec \mathbf{x}^{(b)}$ Dominance relation (solution a dominates b)

$\mathbf{y}_i$ Position of grasp i used in diversity evaluation (see also $\mathbf{y}_i$ in theoretical background)

$\mathbf{y}_i$ Position of the i-th grasp contact point

$\mathcal{A}$ Selected multi-objective evolutionary algorithm (see also $\mathcal{A}$ in multi-suction optimization)

$\mathcal{A}$ Selected multi-objective evolutionary algorithm (see also $\mathcal{A}$ in two-finger optimization)

$\mathcal{B}(\mathbb{R}^k)$ Borel σ-algebra on $\mathbb{R}^k$

$\mathcal{D}$ Observed data set

$\mathcal{D}_t$ Dataset of completed Bayesian optimization trials after t evaluations (see also $\mathcal{D}_t$ in two-finger optimization)

$\mathcal{D}_t$ Dataset of completed Bayesian trials after t evaluations (see also $\mathcal{D}_t$ in multi-suction optimization)

$\mathcal{F}$ Discrete Fourier transform (see also Fourier transform for spatial analysis)

$\mathcal{F}^{(t)}$ Objective values of the population at generation t (see also $\mathcal{K}_{\mathrm{EA}}^{(g)}$, $\mathcal{K}_{\mathrm{BO}}^{(t)}$ in two-finger optimization for similar concepts)

$\mathcal{GP}$ Gaussian process

$\mathcal{G}_{\mathrm{clustered}}$ Reduced clustered grasp set used for diversity evaluation (see also diversity metrics in theoretical background)

$\mathcal{K}_{\mathrm{BO}}^{(t)}$ Non-dominated KPI set after t Bayesian evaluations (see also $\mathcal{F}^{(t)}$ in multi-suction optimization for similar concept)

$\mathcal{K}_{\mathrm{EA}}^{(g)}$ Non-dominated KPI set after evolutionary generation g (see also $\mathcal{F}^{(t)}$ in multi-suction optimization for similar concept)

$\mathcal{M}$ Bayesian surrogate-modeling procedure (see also $\mathcal{M}$ in multi-suction optimization)

$\mathcal{M}$ Bayesian surrogate-modeling procedure (see also $\mathcal{M}$ in two-finger optimization)

$\mathcal{O}(kN^2)$ Computational complexity

$\mathcal{P}^*$ Set of Pareto-optimal solutions in the reduced decision space (see also $\mathcal{P}^*$ for Pareto-optimal solutions)

$\mathcal{P}^{(0)}$ Initial evolutionary population (see also $\mathcal{P}^{(0)}$ in multi-suction optimization)

$\mathcal{P}^{(0)}$ Initial population of candidate decision vectors (see also $\mathcal{P}^{(0)}$ in two-finger optimization)

$\mathcal{P}^*$ Set of Pareto-optimal gripper configurations (see also $\mathcal{P}^*$ for Pareto-optimal solutions)

$\mathcal{Q}$ Admissible search space of decision vectors

$\mathcal{Q}_{t+1}$ Batch of candidates generated by Bayesian optimization (see also $\mathcal{X}_{t+1}$ in two-finger optimization for similar concept)

$\mathcal{X}$ Bounded decision space (see also $\mathcal{X}$ in two-finger optimization)

$\mathcal{X}$ Bounded decision space of the two-finger gripper (see also $\mathcal{X}$ for generic decision space)

$\mathcal{X}$ Feasible decision space

$\mathcal{X}_{t+1}$ Batch of gripper candidates generated by Bayesian optimization (see also $\mathcal{Q}_{t+1}$ in multi-suction optimization for similar concept)

$\mathcal{Z}$ Set of random seeds for meta-optimization (see also $\mathcal{Z}$ for random seeds)

$\mathcal{Z}$ Set of random seeds used for meta-optimization (see also $\mathcal{Z}$ for random seeds)

$\mathrm{HV}(A, \mathbf{r})$ Hypervolume indicator

$\mathrm{HV}_{\mathrm{BO}}^{(t)}$ Bayesian hypervolume after t completed evaluations (see also $\mathrm{HV}^{(t)}$ in multi-suction optimization)

$\mathrm{HV}_{\mathrm{EA}}^{(g)}$ Evolutionary hypervolume after generation g (see also $\mathrm{HV}^{(t)}$ in multi-suction optimization)

$\mu(\mathbf{x})$ Predictive mean

$\sigma^2(\mathbf{x})$ Predictive variance

$\mathrm{HV}^{(t)}$ Hypervolume indicator at generation t (see also $\mathrm{HV}_{\mathrm{EA}}^{(g)}$, $\mathrm{HV}_{\mathrm{BO}}^{(t)}$ in two-finger optimization)

$\widehat{\mathbf{g}}_t$ Probabilistic surrogate approximation after t evaluations (see also $\widehat{\mathbf{g}}_t$ in two-finger optimization)

$\widehat{\mathbf{k}}_t$ Probabilistic surrogate approximation after t evaluations (see also $\widehat{\mathbf{g}}_t$ in multi-suction optimization)

$\zeta_{\mathcal{G},\mathcal{W}}$ Gripper–workpiece collision map (see also collision detection procedures)

A Approximation set

$A(\mathbf{p})$ Area of the triangle formed by the three suction cups (see also triangle area for grasp stability)

A_{seal} Effective sealed area

b Maximum number of Bayesian candidates requested per batch (see also b in multi-suction optimization)

b Maximum number of Bayesian candidates requested per batch (see also b in two-finger optimization)

$B_{\max}$ Maximum number of Bayesian candidate-generation batches (see also $B_{\max}$ in multi-suction optimization)

$B_{\max}$ Maximum number of Bayesian optimization batches (see also $B_{\max}$ in two-finger optimization)

d Number of decision variables

d Number of decision variables

$D(\mathbf{p})$ Diversity metric of the grasp distribution (see also D_{pos}, D_{rot} in theoretical background)

d_{geo} Geodesic distance on $SO(3)$

$d_{\text{geo}}(\mathbf{R}_a,\mathbf{R}_b)$ Geodesic distance between two grasp orientations (see also d_{geo} in theoretical background)

d_{gn} Depth parameter in the common gripper interface

$d_{\text{ori}}(i,j)$ Pairwise orientation distance between grasp poses i and j (see also D_{rot} for orientation diversity)

D_{pos} Normalized spatial diversity

$D_{\text{pos}}(\mathbf{p})$ Spatial component of the diversity metric (see also D_{pos} in theoretical background)

$d_{\text{pos}}(i,j)$ Pairwise translational distance between grasp poses i and j (see also D_{pos} for spatial diversity)

D_{rot} Normalized orientational diversity

$D_{\text{rot}}(\mathbf{p})$ Orientation component of the diversity metric (see also D_{rot} in theoretical background)

E Cumulative number of completed external evaluations (see also $E_{\text{max}}^{\text{BO}}$ for Bayesian evaluation budget)

e_i i-th bounding-box side length

$E_{\max}$ Maximum number of evolutionary evaluations (see also $E_{\max}$ in multi-suction optimization)

$E_{\max}$ Maximum number of evolutionary objective evaluations (see also $E_{\max}$ in two-finger optimization)

$E_{\max}^{\mathrm{BO}}$ Maximum requested number of Bayesian objective evaluations (see also $E_{\max}^{\mathrm{BO}}$ in multi-suction optimization)

$E_{\max}^{\mathrm{BO}}$ Maximum requested number of Bayesian objective evaluations (see also $E_{\max}^{\mathrm{BO}}$ in two-finger optimization)

$f_i(\mathbf{x})$ Individual objective function

F_{hold} Holding force

$f_{\min}$ Best observed objective value

f_{tol} Objective-space convergence tolerance (see also f_{tol} in multi-suction optimization)

f_{tol} Objective-space convergence tolerance (see also f_{tol} in two-finger optimization)

g Evolutionary generation index (see also t in multi-suction optimization)

$g_j(\mathbf{q})$ j-th inequality constraint function

$G_{\max}$ Maximum number of evolutionary generations (see also $G_{\max}$ in multi-suction optimization)

$G_{\max}$ Maximum number of evolutionary generations (see also $G_{\max}$ in two-finger optimization)

h_1,h_2,h_3 Spline contour heights

k Number of objectives

$k_m(\mathbf{x})$ Grasp-quality KPI for objective m (see also k_m for grasp quality metrics)

l_{gn} Length of the gripping contour

m Number of objective functions

m Number of spline interpolation points

$m(\mathbf{x})$ Mean function

N Evolutionary population size (see also N in multi-suction optimization)

N Population size (see also N in two-finger optimization)

N Population size

n Number of grasps in the considered set

n Number of grasps used for diversity evaluation (see also n in theoretical background)

n_c Number of explicit inequality constraints

n_{obs} Number of observations

n_{obj} Number of objectives (see also n_{obj} in multi-suction optimization)

n_{obj} Number of objectives (see also n_{obj} in two-finger optimization)

p_{atm} Atmospheric pressure

$q_i^{\mathrm{L}}, q_i^{\mathrm{U}}$	Lower and upper bounds of the i-th decision variable
q_{intra}	Maximum number of local orientation-diverse representatives (see also clustering procedure)
r	Common radial distance of the suction cups from the gripper center
$r_{\max}$	Upper bound on the common suction cup radius (see also $\mathcal{X}$ for decision space bounds)
$r_{\min}$	Lower bound on the common suction cup radius (see also $\mathcal{X}$ for decision space bounds)
S	Safety factor
s	Characteristic object scale (bounding-box diagonal)
s	Characteristic object scale derived from the mesh bounding box (see also s in theoretical background)
$s_{\mathcal{G}}$	Gripper voxel occupancy grid (see also s for characteristic object scale)
$s_{\mathcal{W}}$	Workpiece voxel occupancy grid (see also s for characteristic object scale)
$SO(3)$	Special orthogonal group of 3D rotations
t	Evolutionary generation index (see also t in two-finger optimization)
$T(\mathbf{q})$	Transformation from reduced decision variables to Cartesian suction cup coordinates
w	Number of generations in the convergence window (see also w in multi-suction optimization)
x_{tol}	Design-space convergence tolerance (see also x_{tol} in multi-suction optimization)

x_{tol} Design-space convergence tolerance (see also x_{tol} in two-finger optimization)

y_i Observed objective value at $\mathbf{x}_i$

z_0 Fixed height coordinate of the suction cups

1 Introduction

1.1 Motivation

Robotic grasping is a fundamental capability in automated manufacturing, logistics, and service robotics[5]. The ability to reliably grasp objects of varying shapes, sizes, and surface properties remains a challenging problem, particularly in unstructured environments[6]. Industrial applications such as bin picking, assembly, and material handling require grippers that can operate robustly across a wide range of workpieces[7].

Among the commonly used end-effectors, suction-cup grippers and two-finger grippers represent two complementary approaches[8]. Suction grippers are widely used for handling objects with smooth surfaces due to their simplicity and speed, whereas two-finger grippers provide flexibility in handling objects with complex geometries. Designing optimal configurations for these grippers is non-trivial, as grasp performance depends on multiple interacting factors such as geometry, contact distribution, force limits, and collision constraints[9].

While the gripper configuration problems are well-motivated in themselves, the primary objective of this thesis is not simply to find the best gripper configuration, but to systematically apply and compare optimization algorithms for solving these configuration problems. A large number of optimization algorithms exist, spanning evolutionary strategies, decomposition-based methods, indicator-based selection, and Bayesian optimization, yet their relative suitability for gripper design problems of this type is not well understood. Different algorithms may exhibit substantially different convergence behaviour, solution quality, and robustness depending on the structure of the objective landscape and the dimensionality of the search space. Identifying which algorithm class performs best for each gripper problem is therefore a practically relevant and scientifically meaningful research question[10].

1.2 Problem Definition

The design of task-specific robotic grippers remains largely a manual and experience-driven process. Geometric parameters, such as the placement of suction cups or the contour of gripper jaws, are commonly selected iteratively and subsequently validated through simulation, geometric evaluation, or physical experiments [11]. This procedure can become time-consuming when many possible configurations must be considered.

For the gripper-design problems investigated in this thesis, the performance of a candidate configuration is determined by computational evaluation routines. These routines do not provide analytical gradients or closed-form expressions that directly relate the design parameters to the resulting performance. Furthermore, a gripper design must generally satisfy several criteria simultaneously, for example by performing across multiple workpieces, grasp locations, or geometric requirements. Improvements with respect to one criterion may reduce performance with respect to another. Representing all criteria by a single aggregated value could hide these trade-offs.

These characteristics lead to expensive, derivative-free, multi-objective black-box optimization problems [12]. A variety of optimization approaches can be applied to such problems, including evolutionary multi-objective algorithms and Bayesian optimization methods. However, their performance depends on factors such as the problem representation, number of objectives, objective landscape, and available evaluation budget [13, 14]. Consequently, results obtained for generic test problems or other applications cannot necessarily be transferred directly to task-specific gripper design.

The primary goal of this thesis is therefore to apply and comparatively evaluate selected evolutionary and Bayesian multi-objective optimization methods for automated gripper configuration. Two application cases are considered: the configuration of a multi-suction-cup gripper and the optimization of rigid jaw contours for a two-finger gripper. In the first case, the positions of the suction cups are optimized to obtain layouts that perform well for a given workpiece. In the second case, spline-based jaw contours are optimized to provide suitable geometric conformity at predefined grasp points. The two cases provide different geometric representations and objective structures through which the behaviour of the selected optimization algorithms can be investigated.

The objective is not to identify a universally superior optimization algorithm. Instead, the thesis examines how the selected methods perform on the two specific gripper-design

problems under comparable experimental conditions. The comparison considers the quality of the obtained multi-objective solutions, their development over the available evaluation budget, and the consistency of the results across repeated runs.

To achieve this goal, the following tasks are defined:

- **Problem formulation:** Formulate the multi-suction and two-finger gripper configuration problems as multi-objective black-box optimization tasks by specifying their decision variables, objectives, and applicable constraints.
- **Framework integration:** Establish a common optimization interface through which the selected algorithms can access both gripper-evaluation pipelines under consistent conditions.
- **Evaluation preparation:** Integrate and adapt the problem-specific evaluation procedures required to assess candidate suction-cup layouts and rigid jaw contours.
- **Experimental design:** Define representative workpieces, predefined evaluation configurations, evaluation budgets, initialization conditions, repeated runs, and performance indicators for a controlled and reproducible comparison.
- **Algorithm application and comparison:** Apply selected evolutionary and Bayesian multi-objective optimization methods to both gripper-design problems and compare their performance using suitable multi-objective indicators.
- **Result analysis:** Analyze the obtained results with respect to solution quality, convergence behaviour, robustness across repeated runs, and sensitivity to the characteristics of the investigated gripper-design problems.

2 Literature Review

2.1 Theoretical Background

2.1.1 Robotics and Object Handling

Robotics

An industrial robot is a automatically controlled, reprogrammable multipurpose manipulator(s), programmable in three or more axes, which can be either fixed in place or fixed to a mobile platform for use in automation applications in an industrial environment[15]. A typical robotic system consists of a kinematic chain of rigid links connected by actuated joints, a set of sensors, a controller and an end-effector mounted at the flange of the last link. The configuration of the manipulator is described by its joint variables, while the position and orientation of the end-effector are described in a Cartesian task space[16]. The mapping between these two spaces is established through forward and inverse kinematics, which form the basis for trajectory planning and motion control[17].

Modern manufacturing systems increasingly require flexibility to accommodate changing production demands, mass customization, increasing product variants, and decreasing batch sizes [18, 19]. Robot-cell reconfiguration and programming nevertheless remain time-consuming and often require specialist expertise [18]. Task-specific assets such as grippers, sensors, and actuators provide robot cells with the capabilities required for particular manufacturing tasks [18].

Bin Picking Problem

Bin picking describes the task of removing individual workpieces from an unordered accumulation of parts, typically supplied in a container, and transferring them to a defined target pose[20, 21]. The task combines perception, grasp planning, collision-free motion planning and physical grasp execution. Because the parts are randomly oriented and may be partially occluded or entangled, the pose of the next object to be grasped is not known in advance and must be estimated from sensor data at runtime[22, 23].

For gripper design, the decisive consequence of this setting is that a grasp cannot be planned for one privileged object pose[20]. The gripper must be able to reach and hold the workpiece across a broad range of orientations and partial accessibility conditions, which implies that a useful gripper configuration provides several alternative grasp candidates distributed over the object surface rather than one single optimal contact[20, 24]. Additional constraints arise from the confined working space inside the bin, where collisions between the gripper body and the bin walls or neighbouring parts restrict the set of feasible approach directions[23, 25]. These requirements directly motivate the objectives used in this thesis, in particular the diversity of the available grasps for the suction cup gripper and the geometric conformance of the finger contour for the two-finger gripper.

2.1.2 Gripper Theory and Geometric Representation

Grasp Theory

A grasp is a configuration of contact points between a gripper and a workpiece such that the gripper can resist external forces and torques applied to the object[26]. The classical criterion for grasp quality is *force closure*. A grasp is force-closed if the contact forces available at the contact points can produce any net wrench on the object, meaning that arbitrary disturbances can be resisted[27]. Force closure requires a sufficient number of contacts with appropriate friction properties and geometric placement.

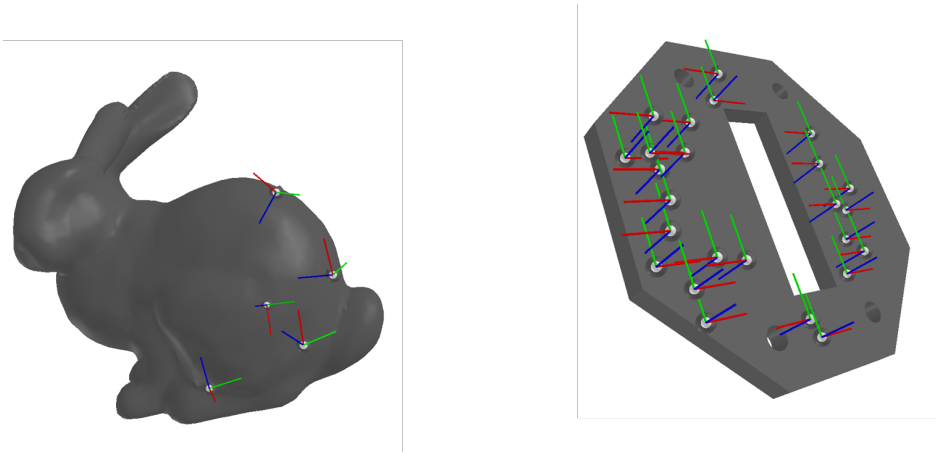


Figure 2.1: Visualisation of a grasp configuration on workpieces.

Figure 2.1 illustrates grasp configurations for two workpieces: the Stanford Bunny and the IPA Intermediate Plate. Each point on the workpiece surface encodes the translation and rotation required to transform the gripper's Tool Center Point (TCP) from its initial pose to the corresponding grasp pose.

Suction Cup Grippers

Suction cup grippers belong to the astrictive gripper class and generate holding force from a pressure difference between the evacuated volume inside the cup and the ambient atmosphere. The holding force is defined

$$F_{\text{hold}} = \frac{\Delta p \cdot A_{\text{seal}}}{S}, \qquad \Delta p \in (0, p_{\text{atm}}] \subset \mathbb{R}_{>0}, \quad A_{\text{seal}} \in \mathbb{R}_{>0}, \quad S \in [1, \infty) \subset \mathbb{R}, \quad F_{\text{hold}} \in \mathbb{R}_{>0} \tag{2.1}$$

where Δp is the pressure difference between the evacuated cup volume and the ambient atmosphere, bounded above by the atmospheric pressure p_{atm} since a perfect vacuum cannot be exceeded, A_{seal} is the effective sealed area enclosed by the cup lip, and $S \geq 1$ is a safety factor accounting for acceleration, orientation and friction conditions[28, 29]. The holding force is strictly positive for any non-vanishing pressure difference acting on a non-degenerate

sealed area. The decisive requirement is therefore an airtight seal between the cup lip and the workpiece surface, which restricts applicable surfaces to regions that are sufficiently smooth, non-porous and locally flat or only moderately curved relative to the compliance of the cup.

Practical implementations frequently use several cups arranged on a common mounting plate. Distributing the load over multiple cups increases the total holding force, provides redundancy if individual cups fail to seal, and, most importantly for handling, generates a resistance to tilting moments[30]. The moment capacity depends on the spatial arrangement of the active cups, which is commonly characterised through the area of the polygon or triangle spanned by the sealing contacts[29]. A large spanned area yields a stable grasp, whereas narrowly clustered cups behave similarly to a single contact. The placement of the cups on the mounting plate is thus the central design variable, and it must be chosen such that valid sealing contacts are achievable on every workpiece of the considered family.

Grasp Diversity

The *diversity* of a grasp set can be quantified by combining two complementary measures of spatial spread.

Geodesic Distance as a Diversity Measure. Orientational diversity is measured by calculating the pairwise geodesic distances between grasp orientations represented by rotation matrices. The geodesic distance describes the minimum angular rotation required to transform one orientation into another, accounting for the geometry of the rotation space. The mean of these pairwise angular distances is normalized by the maximum possible angular distance of 180°, resulting in a diversity score between zero and one. A score close to zero indicates that the grasp orientations are similar, whereas larger values indicate greater average angular separation and thus higher orientational diversity. This measure enables the assessment of how broadly candidate grasps cover different orientations, supporting the selection of diverse grasp configurations.

$$D_{\mathrm{rot}}\big(\{\mathbf{R}_i\}_{i=1}^{n}\big) = \frac{1}{180^\circ}\,\frac{1}{n(n-1)}\sum_{\substack{i,j=1\\ i\neq j}}^{n} d_{\mathrm{geo}}(\mathbf{R}_i,\mathbf{R}_j), \tag{2.2}$$

$$\mathbf{R}_i \in SO(3), \quad d_{\mathrm{geo}} : SO(3)\times SO(3) \to [0^\circ, 180^\circ], \quad D_{\mathrm{rot}} \in [0,1]$$

where $\mathbf{R}_i$ is the rotation matrix of the i-th grasp orientation and d_{geo} the geodesic distance on $SO(3)$, i.e. the minimum angular rotation required to transform one orientation into the other. Normalization by the maximum possible angular distance of 180° yields a score bounded by [0,1], where values close to zero indicate near-identical orientations and larger values indicate greater average angular separation[31, 32].

Spatial Diversity via Mean Pairwise Euclidean Distance. The spatial spread of a set of grasps can be quantified using the mean pairwise Euclidean distance between grasp positions in three-dimensional space. To ensure comparability across objects of different sizes, this distance is normalized by a characteristic object scale s. We define s as the Euclidean norm of the axis-aligned bounding-box side lengths:

$$s = \|\mathbf{e}\|_2, \qquad \mathbf{e} \in \mathbb{R}^3_{>0}, \quad s \in \mathbb{R}_{>0} \tag{2.3}$$

where $\mathbf{e}$ contains the side lengths of the bounding box. The normalized spatial diversity is then defined as

$$D_{\text{pos}}(\{\mathbf{y}_i\}_{i=1}^n) = \frac{1}{s}\,\frac{1}{n(n-1)} \sum_{\substack{i,j=1 \\ i\neq j}}^{n} \|\mathbf{y}_i - \mathbf{y}_j\|_2, \tag{2.4}$$

$$\mathbf{y}_i \in \mathbb{R}^3, \quad i,j \in \{1,\ldots,n\}, \quad n \in \mathbb{N},\, n \geq 2, \quad s \in \mathbb{R}_{>0}, \quad D_{\text{pos}} \in [0,1]$$

where $\mathbf{y}_i \in \mathbb{R}^3$ denotes the position of the i-th grasp contact point and $n \geq 2$ is the number of grasps considered. Normalization by s makes the metric scale-invariant, allowing diversity values to be compared across workpieces of different sizes. Because every pairwise distance is bounded by the bounding-box diagonal, $\|\mathbf{y}_i - \mathbf{y}_j\|_2 \leq s$, the arithmetic mean of these distances is likewise bounded by s and therefore $D_{\text{pos}} \in [0,1]$. The lower limit is attained if and only if all grasp positions coincide, while the upper limit requires every pair of grasps to lie at diagonal distance and is thus only reachable for $n = 2$ at opposite corners of the bounding box. For $n \geq 3$ the attainable values are strictly smaller, so even well-distributed grasp sets are not expected to approach unity. This measure complements the geodesic measure by capturing absolute spatial separation rather than surface-constrained distance.

Multi Finger Grippers

Multi finger grippers are impactive grippers that transmit forces through direct contact between movable jaws and the workpiece. They are classified by the number of fingers, the kinematics of the jaw motion and the number of actuated degrees of freedom[33]. Two-finger parallel grippers with a single actuator represent the industrial standard because of their mechanical simplicity, robustness and low control effort[34]. Three-finger centric designs provide self-centring behaviour for rotationally symmetric parts, while anthropomorphic hands with many degrees of freedom offer high versatility at the cost of complexity.

Two grasp modes are commonly distinguished. Force-closure grasps hold the object through friction at the contact surfaces and therefore depend on the applied clamping force and the friction coefficient. Form-closure grasps immobilise the object through the geometry of the contacts alone, so that no disturbance can move the object without deformation[35]. Because a mechanically simple gripper provides few actuated degrees of freedom, the ability to realise form closure is shifted from the kinematics to the geometry of the jaws. This is the reason why application-specific jaw inserts, frequently produced additively, are widely used in industry and why the jaw contour is the natural design variable for optimization[36].

2.1.3 Optimization Theory

Black-Box Optimization

A black-box optimization problem is one in which the objective function has no known analytical form and no gradient information is available to the optimizer[37]. The function can only be queried at candidate input points, returning a scalar or vector output. This setting arises naturally in engineering design when objectives are evaluated through computationally expensive procedures, making gradient computation either impossible or impractical. Because each function evaluation may be costly, black-box optimizers must find high-quality solutions within a limited evaluation budget, placing emphasis on sample efficiency[38]. The gripper configuration problems addressed in this thesis fall squarely within this category. Objective values are computed through a series of geometric and numerical calculations applied to the candidate gripper configuration, and no closed-form expression exists that directly maps the decision variables to either the contact quality score or the grasp diversity metric.

Multi-Objective Optimization

Multi-objective optimization (MOO) refers to the simultaneous optimization of two or more conflicting objective functions over a shared decision space[39]. Formally, a multi-objective minimization problem is defined as

$$\begin{gathered}\min_{\mathbf{x}\in\mathcal{X}} \mathbf{f}(\mathbf{x}) = \big(f_1(\mathbf{x}), f_2(\mathbf{x}), \ldots, f_k(\mathbf{x})\big), \\ \mathbf{x} \in \mathcal{X} \subseteq \mathbb{R}^d, \quad d \in \mathbb{N}, \quad f_i : \mathcal{X} \to \mathbb{R}, \quad i \in \{1,\ldots,k\}, \quad k \in \mathbb{N}, k \geq 2, \quad \mathbf{f}(\mathbf{x}) \in \mathbb{R}^k\end{gathered} \tag{2.5}$$

where $\mathbf{x}$ is the decision variable vector, $\mathcal{X}$ is the feasible decision space, and $k \geq 2$ is the number of objectives. When objectives conflict, no single solution minimizes all objectives simultaneously. Instead, a set of trade-off solutions exists, collectively forming the Pareto front. The goal of a multi-objective optimizer is therefore not to find a single best solution but to approximate this trade-off surface as accurately and completely as possible[39, 40].

Pareto Optimality

The concept of Pareto optimality provides the theoretical foundation for comparing solutions in a multi-objective setting. A solution $\mathbf{x}^{(a)}$ is said to *dominate* $\mathbf{x}^{(b)}$, written $\mathbf{x}^{(a)} \prec \mathbf{x}^{(b)}$, if and only if

$$\begin{gathered}f_i(\mathbf{x}^{(a)}) \leq f_i(\mathbf{x}^{(b)}) \quad \forall\, i \in \{1,\ldots,k\} \quad \text{and} \quad f_j(\mathbf{x}^{(a)}) < f_j(\mathbf{x}^{(b)}) \quad \text{for at least one } j \in \{1,\ldots,k\}, \\ \mathbf{x}^{(a)}, \mathbf{x}^{(b)} \in \mathcal{X}, \quad k \in \mathbb{N}, k \geq 2\end{gathered} \tag{2.6}$$

A solution is called *Pareto optimal* if no other feasible solution dominates it. The set of all Pareto-optimal solutions in the decision space is the *Pareto set*, and its image in the objective space is the *Pareto front*. In practice, algorithms compute an approximation of the Pareto front, and the quality of this approximation is assessed using performance indicators such as the hypervolume indicator[41, 42].

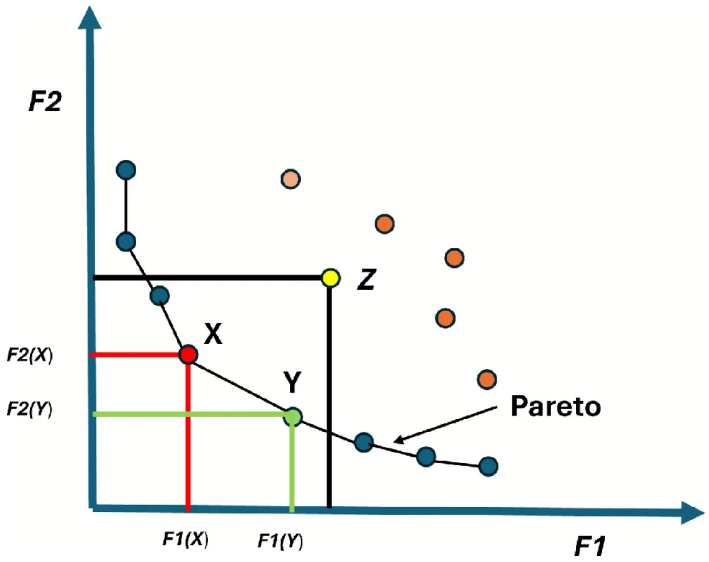


Figure 2.2: Example of a Pareto Front.[1]

Figure 2.2 illustrates the concept of dominated and non-dominated solutions in a multi-objective optimization (MOP) problem with two objectives. The objective is to minimize both objective functions. The connected curve represents the Pareto front (PF), which includes all Pareto-optimal solutions. Solution X_2 has higher values for both f_1 and f_2 when compared to solutions X_1 and Y. Thus, solutions X_1 and Y dominate solution X_2. However, since neither X_1 nor Y dominates the other, they are considered non-dominated solutions[1].

Many-Objective Optimization

Problems with four or more objectives are commonly referred to as many-objective optimization problems[43]. As the number of objectives grows, the proportion of mutually non-dominated solutions in a randomly sampled population increases rapidly, an effect known as dominance resistance. Pareto dominance then loses its ability to discriminate between candidates, so selection pressure towards the Pareto front deteriorates and the search degenerates into an essentially random walk guided mainly by diversity preservation[43, 44].

The number of solutions required to cover a k-dimensional front grows exponentially with k, visual inspection of the approximation set becomes impractical, and the computation of

indicators such as the hypervolume becomes expensive[4]. Algorithm design responds to these issues either by replacing dominance with decomposition into scalar subproblems, by introducing reference points or reference directions to structure the objective space, or by using an indicator as the selection criterion[43, 45].

No Free Lunch Theory

The No Free Lunch (NFL) theorems for optimization state that, averaged over the set of all possible objective functions, all non-repeating search algorithms exhibit identical expected performance[46, 47]. Any advantage that an algorithm demonstrates on one class of problems is therefore compensated by a corresponding disadvantage on another class. Corresponding results have been derived for the multi-objective case[47].

The practical consequence is that there is no universally superior optimization algorithm and that the question of which algorithm to use is only meaningful with respect to a specific problem class[48]. This provides the methodological justification for the empirical approach taken in this thesis. Rather than seeking a generally best algorithm, the objective is to determine which algorithms perform well on the particular structure of gripper configuration problems, and how sensitive this ranking is to workpiece geometry and to the dimensionality of the objective space.

Evolutionary Algorithms

Evolutionary algorithms (EAs) are a family of population-based metaheuristics inspired by the principles of biological evolution. A population of candidate solutions is maintained across discrete generations. At each generation, new candidate solutions are generated through variation operators, typically crossover, which recombines pairs of existing solutions, and mutation, which introduces random perturbations, and a selection mechanism retains the most promising individuals for the next generation based on their objective values. This loop continues until a termination criterion is met, such as a maximum number of function evaluations or convergence of the population[49].

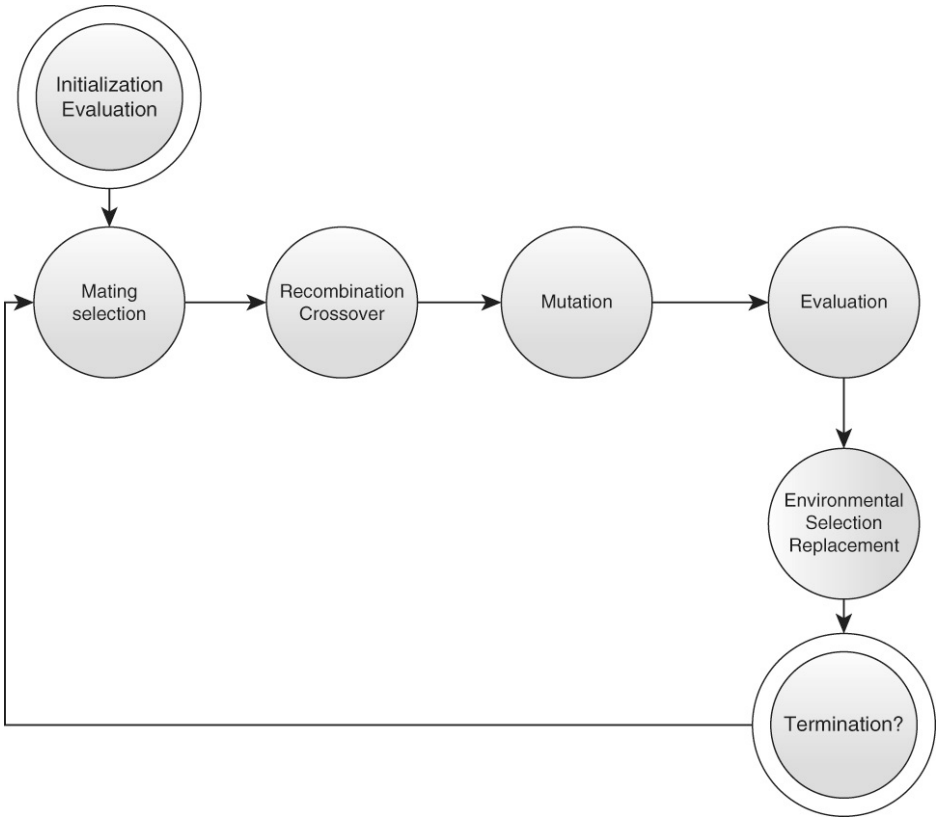


Figure 2.3: The evolutionary cycle, basic working scheme of all EAs [2].

Figure 2.3 shows the basic working scheme shared by all EAs. After initialization and evaluation of the initial population, the algorithm iteratively applies mating selection, recombination (crossover), mutation, evaluation of the offspring, and environmental selection (replacement), until a termination criterion is satisfied

EAs are well suited to black-box multi-objective optimization because they require no gradient information, naturally maintain a diverse population of trade-off solutions, and are robust to noisy or discontinuous objective landscapes. Multi-objective evolutionary algorithms (MOEAs) extend the basic EA framework with selection mechanisms based on Pareto dominance or decomposition, enabling them to approximate the Pareto front in a single run[50, 51].

Bayesian Optimization

Bayesian optimization (BO) is a sequential model-based strategy for the global optimization of expensive black-box functions. A probabilistic surrogate model, most commonly a Gaussian process (GP), is fitted to the observations collected so far[52]. The surrogate provides both a mean prediction and an uncertainty estimate at unobserved locations, which together define an *acquisition function* that balances exploration of uncertain regions with exploitation of promising ones[52, 53]. The point that maximizes the acquisition function is selected as the next evaluation, after which the surrogate is updated and the process repeats[54][3]. Because the surrogate concentrates the sampling effort on informative regions, BO typically requires far fewer function evaluations than population-based methods, which makes it attractive whenever the evaluation budget is the limiting factor.

Gaussian Process Surrogates. A Gaussian process defines a distribution over functions and is fully specified by a mean function $m(\mathbf{x})$ and a positive semi-definite covariance or kernel function $\kappa(\mathbf{x}, \mathbf{x}')$, written as $f(\mathbf{x}) \sim \mathcal{GP}\big(m(\mathbf{x}), \kappa(\mathbf{x}, \mathbf{x}')\big)$. Conditioning the prior on the observed data set $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{n_{\text{obs}}}$, where $y_i \in \mathbb{R}$ is the response measured at $\mathbf{x}_i$ and n_{obs} the number of evaluations performed so far, yields a posterior that is again Gaussian, with closed-form predictive mean $\mu(\mathbf{x})$ and non-negative predictive variance $\sigma^2(\mathbf{x})$ at any query point[55]. The kernel encodes the assumed smoothness and length scale of the underlying function, with the Matérn and squared exponential kernels being the most common choices, and its hyperparameters are estimated by maximizing the marginal likelihood[55]. An additional noise term accounts for stochastic or numerically imprecise evaluations[56]. The main limitations of GP surrogates are the cubic scaling of exact inference with the number of observations and a loss of predictive quality in high-dimensional decision spaces[52].

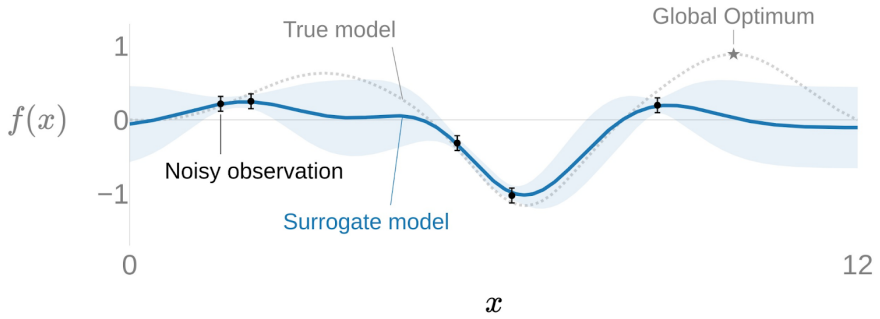


Figure 2.4: Surrogate Model [3].

In figure 2.4 a surrogate model is fit to five noisy observations using a GP to predict the objective, depicted by the solid line, and uncertainty estimates, illustrated by the width of the shaded bands. This objective is predicted for the entire range of possible parameter values, corresponding to the full x-axis. Importantly, the model is able to predict the outcome and quantify the uncertainty of configurations that have not yet been tested. Intuitively, the uncertainty bands are tight in regions that are well-explored and become wider as we move away from them.

Acquisition Functions. The acquisition function converts the posterior of the surrogate into a scalar utility that quantifies how promising a candidate point is. Expected improvement (EI) measures the expected amount by which a candidate improves on the best observed value,

$$\begin{aligned} \alpha_{\mathrm{EI}}(\mathbf{x}) &= \mathbb{E}\big[\max\big(f_{\min} - f(\mathbf{x}), 0\big)\big], \\ \mathbf{x} \in \mathcal{X} \subseteq \mathbb{R}^d, \quad f_{\min} &= \min_{i \in \{1,\dots,n_{\mathrm{obs}}\}} y_i \in \mathbb{R}, \quad \alpha_{\mathrm{EI}} : \mathcal{X} \to \mathbb{R}_{\geq 0} \end{aligned} \tag{2.7}$$

where $\mathbb{E}[\,\cdot\,]$ denotes the expectation taken with respect to the posterior of the surrogate at $\mathbf{x}$ and $f_{\min}$ is the best objective value observed so far. Truncating the improvement at zero renders the acquisition non-negative, so that candidates which are not expected to improve on the incumbent receive zero utility. The expression admits a closed form for a Gaussian posterior[56, 3].

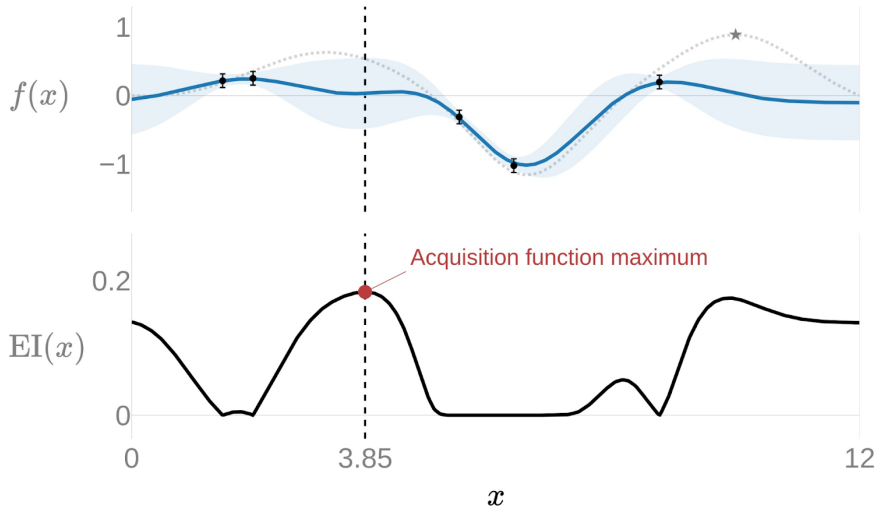


Figure 2.5: Acquisition Function [3].

A visualization of the expected improvement based on the surrogate model predictions is shown in Figure 2.5, where the next suggestion is where the expected improvement is at its maximum. Once a new highest EI is selected and evaluated, the surrogate model is retrained and a new suggestion is made. This process continues iteratively until a stopping condition, set by the user, is reached [3].

Since the acquisition function is inexpensive to evaluate and analytically differentiable, its maximization is performed by an inner optimizer rather than by querying the expensive objective.

Multi-Objective Bayesian Optimization. Extending BO to several objectives requires a utility notion for sets instead of single points. One family of methods fits an independent surrogate per objective and scalarizes the problem with randomly drawn weight vectors, so that the front is sampled gradually over successive iterations[57]. A second and now dominant family uses the hypervolume as the underlying utility[58]. Expected hypervolume improvement and its noisy, batch-capable variants select the candidates that are expected to enlarge the dominated hypervolume the most, which aligns the acquisition criterion directly with the indicator used for evaluation[57]. Information-theoretic approaches instead maximize the information gained about the Pareto front. Multi-objective BO is particularly appealing under tight evaluation budgets, but its cost per iteration grows with the number of objectives

and observations, which makes the comparison against evolutionary methods on the gripper problems a question of the available budget rather than of asymptotic quality alone.

Performance Indicators

Because a multi-objective optimizer returns a set rather than a single point, comparing algorithms requires indicators that map an approximation set to a scalar quality value[4]. Indicators are usually grouped according to whether they assess convergence towards the front, the diversity of the solutions along the front, or both properties at once.

The *hypervolume* indicator (HV) measures the volume of the objective space that is dominated by the approximation set and bounded by a reference point $\mathbf{r}$,

$$\begin{gathered} \mathrm{HV}(A, \mathbf{r}) = \lambda\Big(\bigcup_{\mathbf{a} \in A} [\mathbf{f}(\mathbf{a}), \mathbf{r}] \Big), \\ A \subset \mathcal{X},\ |A| < \infty, \quad \mathbf{r} \in \mathbb{R}^k \text{ with } f_i(\mathbf{a}) \le r_i \ \forall\, \mathbf{a} \in A, i \in \{1, \ldots, k\}, \\ \lambda : \mathcal{B}(\mathbb{R}^k) \to \mathbb{R}_{\ge 0} \text{ (Lebesgue measure)}, \quad \mathrm{HV} \in \mathbb{R}_{\ge 0} \end{gathered} \tag{2.8}$$

where λ denotes the Lebesgue measure on the Borel sets $\mathcal{B}(\mathbb{R}^k)$ and A is the finite approximation set returned by the optimizer[41, 59]. The reference point $\mathbf{r}$ must be dominated by every member of A, that is $f_i(\mathbf{a}) \le r_i$ for all $\mathbf{a} \in A$ and all i, so that each box $[\mathbf{f}(\mathbf{a}), \mathbf{r}]$ is non-empty and the indicator is well defined and non-negative. The hypervolume is the most widely used indicator because it is Pareto compliant, meaning that a strictly dominating set never receives a worse value, and because it requires no knowledge of the true Pareto front[4, 41].

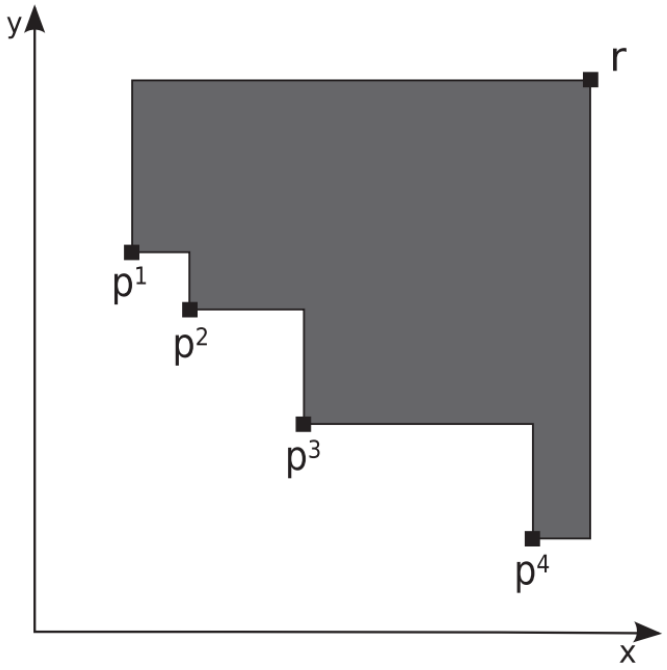


Figure 2.6: Hypervolume indicator [4].

Figure 2.6 shows a two-dimensional example of the hypervolume (an area) and a fixed reference point r.

Inverted generational distance. The *inverted generational distance* (IGD) measures the mean distance from each point of a reference set Z to the closest point of the approximation set and therefore requires a reference front, which is normally unavailable for real-world black-box problems[59]. IGD is furthermore not Pareto compliant: a set that dominates another can receive a worse indicator value[59]. The modified variant IGD^+ removes this defect by replacing the Euclidean distance with a dominance-aware distance that only penalizes those objective components in which the approximation set is actually worse than the reference point[60].

$$\mathrm{IGD}^+(A, Z) = \frac{1}{|Z|} \sum_{\mathbf{z} \in Z} \min_{\mathbf{a} \in A} \sqrt{\sum_{i=1}^{k} \big(\max\{\, a_i - z_i, 0 \,\}\big)^2}, \qquad \mathrm{IGD}^+ \in \mathbb{R}_{\geq 0}, \tag{2.9}$$

where A is the approximation set under evaluation and a_i and z_i denote the i-th objective of an approximation and a reference point, respectively. Lower values indicate better approximation quality, and IGD^+ is weakly Pareto compliant[60]. Because the true Pareto front of the gripper design problem is unknown, the reference set Z is constructed as the non-dominated union of the fronts of all compared runs[59]. IGD^+ is therefore a *relative* measure: it quantifies how well an individual run approximates the best front jointly attainable by the whole set of runs, and it is optimistically biased in favour of those runs that contribute many points to Z[59].

2.1.4 Multi-Objective Evolutionary Algorithms

The candidate algorithms evaluated in this thesis are all members of the multi-objective evolutionary algorithm (MOEA) family. Each follows the general EA loop described in Section 2.1.3 but differs in how it guides selection toward a well-distributed approximation of the Pareto front. Brief introductions to each algorithm are given below.

NSGA-II

The Non-dominated Sorting Genetic Algorithm II (NSGA-II) is among the most widely adopted MOEAs. Its selection mechanism is based on two criteria applied in sequence. First, the entire population is sorted into non-domination ranks, so that all non-dominated solutions receive rank 1, they are removed, and the process repeats until every individual has a rank. Second, within the same rank, solutions are ordered by *crowding distance*, a density estimate that measures how isolated a solution is from its neighbours in the objective space. Solutions with a higher crowding distance are preferred, encouraging spread along the Pareto front. NSGA-II uses an elitist strategy in which the combined parent and offspring population of size $2N$ is truncated to N using these two criteria. Its computational complexity per generation is $\mathcal{O}(kN^2)$, where k is the number of objectives and N is the population size[61].

NSGA-III

NSGA-III extends NSGA-II to problems with three or more objectives (many-objective problems), where crowding distance becomes an unreliable diversity measure. Instead of crowd-

ing distance, NSGA-III uses a set of structured *reference points* distributed uniformly on a normalized hyperplane in the objective space. Solutions are associated with their closest reference point, and selection preference is given to solutions that are associated with under-represented reference points and lie closest to their reference line. This mechanism explicitly promotes diversity across the objective space and has been shown to outperform NSGA-II on problems with three or more objectives[62].

MOEA/D

MOEA/D (Multi-Objective Evolutionary Algorithm based on Decomposition) takes a fundamentally different approach from Pareto-ranking methods. The multi-objective problem is decomposed into N scalar subproblems using a set of evenly distributed weight vectors and an aggregation function such as the weighted Chebyshev scalarization. Each subproblem is optimized simultaneously, with solutions sharing information within a defined neighbourhood of weight vectors. This neighbourhood-based collaborative search makes MOEA/D computationally efficient and well suited to problems where the Pareto front has a regular structure. Its complexity per generation is lower than NSGA-II, as it avoids global non-dominated sorting[63, 64].

SMS-EMOA

The S-Metric Selection Evolutionary Multi-Objective Algorithm (SMS-EMOA) uses the hypervolume indicator directly as its selection criterion. It operates as a steady-state algorithm, so that at each step a single offspring is generated, added to the population, and the individual that contributes the least to the population hypervolume is removed. This ensures that the population always converges toward solutions that maximize the dominated hypervolume with respect to a reference point. SMS-EMOA is particularly well suited to problems where solution quality is to be measured by the hypervolume indicator, as its selection pressure is directly aligned with that metric[65].

CMOPSO

The Competitive Multi-Objective Particle Swarm Optimizer (CMOPSO) adapts the particle swarm optimization (PSO) framework to the multi-objective setting. In standard PSO, each

particle moves through the decision space guided by its personal best position and a global best position. CMOPSO introduces a competitive mechanism in which particles compete in pairs within the swarm, and the losing particle updates its velocity by learning from the winning particle rather than from a fixed global best. An external archive of non-dominated solutions is maintained to provide guidance and preserve diversity across the Pareto front. This competitive selection strategy avoids premature convergence to a single region of the Pareto front and has been shown to achieve fast convergence while remaining competitive with dominance-based MOEAs on standard benchmark problems[66].

Suction Cup Grippers

2.2 State of the Art

This section compares previous studies according to their optimization target, the gripper properties treated as fixed, and their relevance to task-specific gripper configuration. The reviewed literature can be divided into four groups: grasp planning with fixed end-effectors, multi-suction grasping, two-finger gripper design, and comparative studies of multi-objective optimization algorithms.

2.2.1 Grasp Planning with Fixed End Effectors

Most robotic grasping research treats the end-effector geometry as a fixed input and searches for a feasible grasp pose on the workpiece. Zurbrügg et al. [67], for example, optimize grasp selection while assuming that the geometry and contact configuration of the gripper have already been defined. Analytical grasp-planning methods evaluate candidate contacts using geometric, force-closure, or form-closure conditions, whereas data-driven methods estimate grasp quality from training data. Despite their methodological differences, both approaches primarily vary the grasp pose rather than the gripper geometry.

Although Todescato et al. [5] establish gripper geometry as an optimization variable, their formulation is limited to a single suction cup. Their objective-function evaluation is methodologically similar to the approach used in this thesis, as suction-contact feasibility is assessed using quasi-static criteria related to the methods presented by Kleeberger et al. [68] and the compliant suction-contact model of Mahler et al. [69]. However, their formulation primarily

considers the availability of valid suction grasps and does not explicitly evaluate the distribution/coverage of the resulting grasp set. The present thesis extends this approach to a multi-suction-cup layout by optimizing the spatial positions of the individual cups and evaluating valid grasps using orientation and spatial diversity metrics. These metrics distinguish between layouts that provide a similar number of valid grasps but differ in how broadly those grasps are distributed across positions and orientations.

The studies therefore address complementary but different questions. Grasp planning determines how an existing gripper should interact with a workpiece, whereas gripper-configuration optimization determines which geometry should be available before grasp planning is performed. Improving the grasp planner cannot fully compensate for a suction layout that cannot establish sufficient seals or for a jaw contour that cannot reach suitable contact regions. Consequently, fixed-gripper grasp-planning results do not directly resolve the end-effector design problem considered in this thesis.

2.2.2 Comparison of Multi-Suction Grasping Studies

Existing studies on multi-suction grippers address grasp prediction, experimental stability assessment, and load-aware motion planning, but generally assume predefined suction-cup layouts. Schillinger et al. [70] predict grasp quality from RGB-D images and optimize the grasp pose and gripper selection from a predefined set of multi-suction grippers. Their method selects among existing layouts rather than optimizing the positions of the individual cups.

Geng et al. [71] experimentally investigate how suction-cup size, geometric arrangement, centre of gravity offset, and robot acceleration influence grasp stability. Although different arrangements are evaluated, these are predefined experimental configurations rather than layouts generated through multi-objective optimization.

Lee et al. [72] model the load distribution among multiple suction cups using a quadratic program. They incorporate the resulting grasp-failure constraints into time-optimal trajectory planning to support the fast and reliable manipulation of heavy objects. Their work focuses on load prediction and motion planning for an existing gripper configuration, not on optimizing the cup arrangement.

These studies therefore address complementary aspects of multi-suction grasping. Schillinger et al. focus on grasp-pose and gripper selection, Geng et al. examine the effects of gripper configuration and handling conditions experimentally, and Lee et al. model load distribution during dynamic manipulation. However, none of these studies formulates the positions of the individual suction cups as decision variables in a multi-objective layout-optimization problem.

2.2.3 Comparison of Two-Finger Gripper Studies

Research on two-finger gripper design includes the optimization of compliant fingers, underactuated mechanisms, and task-specific jaw geometries. Sun et al. [73] use three-dimensional topology optimization to design compliant continuum fingers and a lightweight gripper base. The resulting fingers deform passively to adapt to objects with different shapes. Similarly, Xie et al. [74] optimize the topology of flexible fingers while accounting for contact behaviour and contact-stress constraints. Their objective is to achieve adaptive, damage-free grasping of fragile and geometrically variable objects. Both studies focus on structural compliance and deformation, whereas this thesis considers the geometric optimization of rigid jaw contours.

Dong et al. [75] optimize the geometry and transmission system of an underactuated tendon-driven gripper. Their design variables include phalange dimensions, palm width, joint-mandrel radii, and tendon-routing parameters. The optimization aims to improve force transmission and grasp stability. Although this approach optimizes gripper geometry, it concerns the dimensions and actuation mechanism of an articulated gripper rather than the contact contour of a rigid parallel jaw.

Reichert [76] developed the direct predecessor of the parallel-jaw optimization framework continued in this thesis. The work represents the contact contour of a rigid parallel jaw using a parametric spline and evaluates candidate contours through a collision-map-based procedure. Intersections close to the intended workpiece boundary contribute positively to a geometric key performance indicator (KPI), whereas intersections extending into the workpiece interior are penalized. The KPI therefore measures the geometric conformity of the generated jaw contour at a specified evaluation configuration rather than providing a complete force- or wrench-based assessment of grasp stability.

Reichert formulates the jaw-design problem using two fixed workpiece orientations, each contributing an objective to the optimization. This demonstrates the feasibility of combining spline-based jaw generation, geometric KPI evaluation, and multi-objective black-box optimization. However, the formulation is restricted to the two investigated evaluation configurations.

This thesis continues Reichert's work by extending the evaluation from two fixed configurations to a set of n predefined grasp points. The grasp points remain fixed and are not optimization variables; the decision variables are the parameters defining the spline-based jaw contour. Each candidate contour is evaluated at every predefined grasp point, producing one geometric KPI for each point. All n KPIs are retained as separate objectives and maximized simultaneously, resulting in an n-objective optimization problem with an n-dimensional objective space.

The optimization therefore seeks Pareto-optimal rigid jaw contours that provide different compromises in geometric conformity across the predefined grasp points. Compared with the studies on compliant and underactuated grippers, the formulation concerns rigid task-specific contact contours. Compared with Reichert's direct predecessor formulation, the principal extension is the evaluation and optimization of one separate objective for each of the n predefined grasp points.

2.2.4 Comparative Studies of Multi-Objective Optimization Algorithms

Comparative studies show that optimizer performance depends strongly on the problem and experimental setup. Tanabe and Ishibuchi [77] evaluate evolutionary algorithms on a suite of real-world multi-objective problems with varied characteristics, while Tian et al. [78] propose selecting algorithms based on problem features. Ishibuchi et al. [79] further demonstrate that termination conditions, population sizes, performance indicators, and test problems can change algorithm rankings. Audet et al. [41] emphasize that indicators assess different properties of Pareto-front approximations, such as convergence, distribution, and spread.

For expensive optimization, Qin et al. [80] compare two combinations of evolutionary and Bayesian optimization on ZDT and DTLZ problems and find that Bayesian evolutionary optimization is more efficient under their experimental conditions. Santoni et al. [13] compare high-dimensional Bayesian optimization methods on single-objective BBOB functions

and report that performance varies with dimensionality, function landscape, and evaluation budget. These findings are therefore specific to the investigated problems and settings.

Such results cannot be transferred directly to the collision-aware suction-layout and spline-based jaw-contour problems considered in this thesis. Therefore, this thesis compares selected multi-objective evolutionary and Bayesian algorithms under common evaluation budgets, initialization conditions, repeated runs, and statistical procedures. The aim of this comparison is to identify methods suitable for these specific gripper-design problems rather than a universally superior optimizer.

2.2.5 Comparative Synthesis and Research Gap

The reviewed literature reveals three related research gaps. First, grasp-planning studies generally optimize grasp poses for predefined end-effectors rather than the gripper configuration itself. Second, existing multi-suction studies generally assume predefined cup layouts. For parallel-jaw design, Reichert [76] provides the spline-based formulation used in this thesis but considers only two fixed evaluation configurations. Third, existing algorithm comparisons do not determine which optimization methods are best suited to the two geometric gripper-design problems considered in this thesis.

For the multi-suction gripper, the unresolved problem is to optimize the positions of the individual cups to maximize grasp coverage on every workpiece and the area formed by the cup arrangement. For the two-finger gripper, this thesis continues Reichert's framework by optimizing a rigid spline-based jaw contour across a predefined grasp points. Each grasp point contributes an individual geometric conformity objective. Both formulations are expensive multi-objective black-box problems, but their different representations, numbers of objectives, and evaluation procedures may favour different optimization methods.

Within the literature reviewed for this thesis, no controlled comparison of evolutionary and Bayesian multi-objective algorithms was identified for these two gripper-design formulations.

2.2.6 Research Positioning and Contribution

This thesis addresses these gaps by treating gripper geometry as the primary optimization variable. The multi-suction formulation optimizes the spatial arrangement of individual suction cups, whereas the two-finger formulation optimizes spline-based contours for rigid parallel jaws. Candidate designs are assessed using problem-specific geometric and collision-aware evaluations.

A modular framework separates geometric evaluation from optimization through a common interface. This allows the applicable algorithms to operate on the same problem formulation, constraints, and evaluation conditions. Selected evolutionary and Bayesian multi-objective algorithms are compared under defined evaluation budgets using repeated independent runs, convergence histories, multi-objective performance indicators, and statistical tests. Evaluation counts and optimizer overhead are reported separately where computational cost is relevant.

The contribution of this thesis is therefore a problem-specific comparison rather than a claim of universal algorithm superiority. The results provide evidence for selecting optimization methods for the suction-layout and rigid-jaw-contour formulations considered here.

3 Methodology

This chapter presents the methodological foundations of the gripper optimization study. It describes the research design, problem formulation, geometric parameterizations, objective criteria, selected optimization frameworks and algorithms, workpiece selection, evaluation architecture, and performance indicators. Implementation details and the specific experimental procedure are presented separately in the approach chapter.

3.1 Research Design

3.1.1 Research Paradigm

This thesis adopts a computational experimental methodology. Gripper configurations are represented mathematically, evaluated using computational geometry methods, and optimized through controlled black-box optimization experiments. This methodology is appropriate because the quality of a gripper configuration depends on geometric interactions between the gripper and the workpiece that cannot be expressed conveniently in closed analytical form.

The principal methodological contribution is not the development of a new optimization algorithm. Instead, the thesis provides a controlled comparison of established evolutionary and Bayesian optimization methods under common problem representations, objective definitions, constraints, evaluation interfaces, and computational budgets.

3.1.2 Study Structure

The suction-cup and parallel-jaw grippers are treated as separate optimization studies. Each optimization run considers exactly one workpiece. Different workpieces therefore define separate problem instances rather than simultaneous objectives within the same optimization problem.

For the suction-cup gripper, the positions of three suction cups are optimized for the selected workpiece. The optimization considers the spatial and orientational coverage of the resulting valid grasps and the area of the triangle formed by the three suction cups.

For the parallel-jaw gripper, a spline-based jaw contour is optimized for a predefined set of grasps on the selected workpiece. The contour is evaluated according to how well it conforms to the workpiece geometry associated with the complete predefined grasp set while avoiding geometrically infeasible contact conditions.

Because the two gripper studies use different geometric representations and evaluation criteria, their optimization results are analyzed separately and are not combined into a common Pareto front.

3.1.3 Experimental Setup

Each combination of gripper type and workpiece defines a separate optimization instance. The selected algorithms are applied independently to each instance under comparable conditions. Multiple independent repetitions allow the variability of the stochastic optimization methods to be examined. Within an instance, the algorithms use the same decision-variable bounds, objective definitions, feasibility conditions, and evaluation budget.

3.1.4 Scope and Limitations

The conclusions of the study are limited to the selected gripper parameterizations, workpieces, objective criteria, and evaluation budgets. Although repeated computational runs allow performance variability to be examined, the results do not establish that one algorithm is universally superior across all gripper-design problems.

The evaluation is entirely computational. Manufacturing effects, material properties, contact forces, suction behavior, and experimental validation on physical grippers are outside the scope of the thesis. Consequently, the optimized configurations are evaluated as geometric designs rather than as fully validated physical grippers.

3.2 Problem Formulation

3.2.1 Black-Box Optimization Framework

The gripper-design problems are formulated as constrained black-box optimization problems. Let $\mathbf{q} \in \mathcal{Q}$ denote a candidate gripper configuration and let d denote the number of decision variables. The general vector-valued optimization problem is written as

$$\min_{\mathbf{q}\in\mathcal{Q}} \mathbf{f}(\mathbf{q}) = (f_1(\mathbf{q}), f_2(\mathbf{q}), \ldots, f_m(\mathbf{q}))\,, \tag{3.1}$$

subject to the inequality constraints

$$g_j(\mathbf{q}) \leq 0, \qquad j = 1, \ldots, n_c, \tag{3.2}$$

and the variable bounds

$$q_i^{\mathrm{L}} \leq q_i \leq q_i^{\mathrm{U}}, \qquad i = 1, \ldots, d. \tag{3.3}$$

Objectives that are naturally maximized are transformed to maintain a common minimization convention. For problems with conflicting criteria, a multi-objective formulation preserves alternative compromise solutions, while weighted scalarization collapses the problem to a single solution defined by chosen weights[81].

The number of objectives is determined by the performance criteria of the corresponding gripper problem. It is not determined by the number of workpieces included in the overall study. Each workpiece is optimized independently.

3.2.2 Parameterization of Gripper Geometry

A central methodological decision is the representation of gripper geometry as a finite-dimensional decision vector. The representation must provide sufficient geometric flexibility while remaining compact, interpretable, and suitable for black-box optimization [82, 5, 83].

Suction-Cup Gripper Representation

For the suction-cup gripper, representing the planar Cartesian coordinates of three cups independently would require six decision variables. A reduced polar parameterization is therefore selected. The position of cup i is expressed as

$$\mathbf{p}_i = \begin{pmatrix} r\cos\alpha_i \\ r\sin\alpha_i \end{pmatrix}, \qquad i = 1,2,3, \tag{3.4}$$

which gives the decision vector

$$\mathbf{q}_{\mathrm{SC}} = (r,\alpha_1,\alpha_2,\alpha_3)\,. \tag{3.5}$$

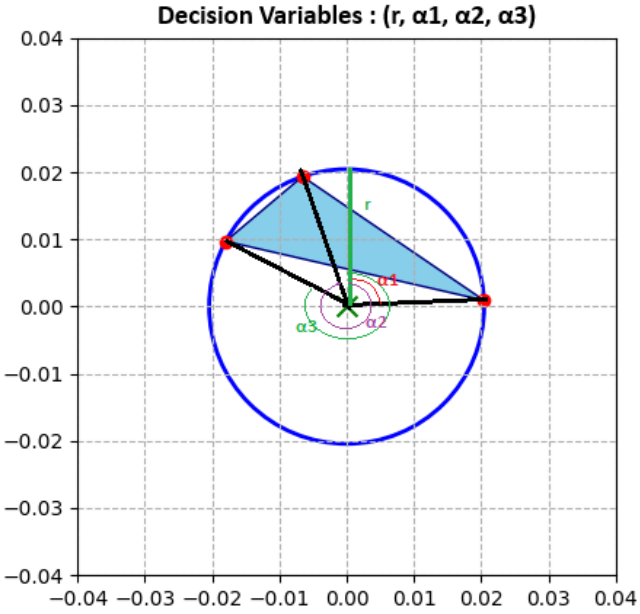


Figure 3.1: polar parameterization of Suction cup gripper.

Figure 3.1 shows the polar coordinates of a potential suction cup gripper. The common radius r determines the distance of the cups from the gripper origin, while the angles α_i determine their arrangement. This representation reduces the dimensionality of the search space and restricts the cup centers to a common circle. Separation constraints are required to prevent overlapping or degenerate cup arrangements.

Parallel-Jaw Gripper and Contour Representation

The parallel-jaw gripper applies gripping forces through two opposing jaw surfaces. The geometry of these surfaces determines how closely the gripper conforms to the workpiece at the predefined grasps. The jaw contour therefore requires a representation that is compact enough for optimization while remaining expressive enough to describe smooth and physically realizable shapes [9].

A cubic interpolating spline is used to represent the jaw contour. Such a spline is a piecewise polynomial curve of degree at most three that passes through a prescribed set of interpolation

points. Adjacent polynomial segments are connected so that the curve and its first and second derivatives remain continuous. This produces a smooth contour, while appropriate endpoint conditions determine the behavior of the spline boundaries.

Spline parameterization reduces the continuous shape-design problem to a finite-dimensional optimization problem. The endpoint interpolation points are fixed to preserve the required spatial extent of the jaw, while three interior interpolation points are variable and define the contour decision variables. Increasing the number of variable points would allow more complex contours but would also increase the dimensionality of the optimization problem [84, 85].

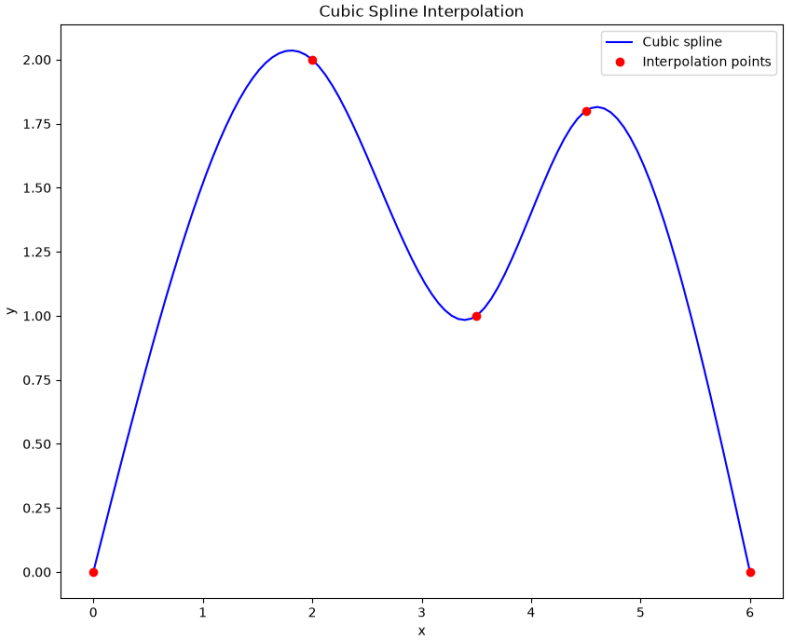


Figure 3.2: Cubic spline interpolation through prescribed interpolation points.

Figure 3.2 illustrates a cubic interpolating spline passing through its prescribed data points. In contrast to a B-spline control-point representation, the interpolation points lie directly on the resulting curve. Consequently, the three variable interior points provide intuitive geometric parameters for modifying the contour, while the fixed endpoints preserve its prescribed spatial extent.

3.2.3 Selection of Objective Functions

The objective functions define the properties used to distinguish effective gripper configurations. They must represent meaningful geometric performance criteria while remaining computable through the available evaluation models [5, 86].

For the suction-cup gripper, the first criterion is the area of the triangle formed by the three cup centers. It is defined as

$$A(\mathbf{q}_{\mathrm{SC}}) = \frac{1}{2} \left| (\mathbf{p}_2 - \mathbf{p}_1) \times (\mathbf{p}_3 - \mathbf{p}_1) \right| . \tag{3.6}$$

The triangle area is used as a geometric measure of the spatial spread of the support layout. A larger area represents a less concentrated arrangement of the suction cups and may improve resistance to rotational disturbances. However, it is a geometric proxy rather than a complete physical stability measure because suction forces, sealing conditions, surface normals, material properties, and the workpiece center of mass are not explicitly modeled [87].

The second suction-cup criterion concerns the spatial and orientational coverage of valid grasps on the selected workpiece. A simple count of feasible grasps does not distinguish between diverse grasps and many nearly identical grasps. Coverage-based evaluation therefore considers the distribution of the available grasps in both position and orientation, providing a measure of the operational flexibility of the cup layout [88, 89].

For the parallel-jaw gripper, the objective criterion evaluates the conformity of a candidate spline contour to a predefined set of grasps on one workpiece. The contour should provide close geometric correspondence with the workpiece across the complete grasp set rather than being specialized for only one grasp. A voxel-based convolution method developed at Fraunhofer IPA is used to assess finger-to-workpiece proximity and contact quality while accounting for geometrically infeasible or colliding configurations [82, 90].

3.3 Optimization Library Selection

The evolutionary optimization algorithms are selected from the `pymoo` framework. `pymoo` provides a unified Python interface for multi-objective evolutionary optimization, constraint handling, performance evaluation, and visualization. Using algorithms from a common framework reduces implementation differences and improves the reproducibility of the comparison. The framework also provides established multi-objective performance indicators, including hypervolume and inverted generational distance plus [64].

Bayesian multi-objective optimization is represented using the `Ax` platform and its underlying `BoTorch` functionality. Bayesian optimization constructs probabilistic surrogate models of the black-box objective functions and uses an acquisition function to select promising candidate evaluations. `Ax` provides the experiment-management interface, while `BoTorch` provides the probabilistic modeling and acquisition-function capabilities [3, 91].

The use of established open-source frameworks avoids the need to implement the optimization methods, surrogate models, and acquisition functions from scratch. It also supports a more reproducible comparison between evolutionary and Bayesian optimization strategies.

3.4 Algorithm Selection

For the experimental comparison, NSGA-II, NSGA-III, MOEA/D, SMS-EMOA, CMOPSO, and multi-objective Bayesian optimization are selected.

The selection was made to compare different approaches to multi-objective search under the same gripper-design problems and evaluation budget. Including algorithms with different search principles reduces the extent to which the comparison is limited to variations of a single optimization strategy. It also allows the experiments to determine whether the relative performance depends on the selected gripper type or workpiece.

- **NSGA-II** is selected as the baseline algorithm for the population-based comparison.
- **NSGA-III** is included to compare the NSGA-II baseline with an alternative that organizes the search using predefined reference directions.

- **MOEA/D** is selected to include a decomposition-based approach and to examine how this formulation performs on the objective trade-offs of the gripper-design problems.
- **SMS-EMOA** is included as an indicator-based approach. Its selection also permits investigation of an algorithm whose search is directly associated with hypervolume, one of the performance indicators used in this thesis.
- **CMOPSO** is selected to include a swarm-based search method and thereby extend the population-based comparison beyond evolutionary algorithms.
- **Multi-objective Bayesian optimization** is included as a model-based alternative to the population-based methods. This allows its performance under a limited evaluation budget to be compared with that of the other selected algorithms.

Together, these algorithms form a deliberately varied comparison set containing dominance-based, reference-direction-based, decomposition-based, indicator-based, swarm-based, and model-based approaches. The experiments use this selection to determine which approaches provide suitable convergence, trade-off representation, and consistency for the suction-cup and parallel-jaw gripper-design problems.

3.5 Workpiece Selection

The workpieces are selected to provide geometrically different problem instances rather than to optimize the grippers for one particular object. Geometric diversity is important because algorithm performance can depend on the structure of the feasible design space and the resulting optimization landscape [92, 93].

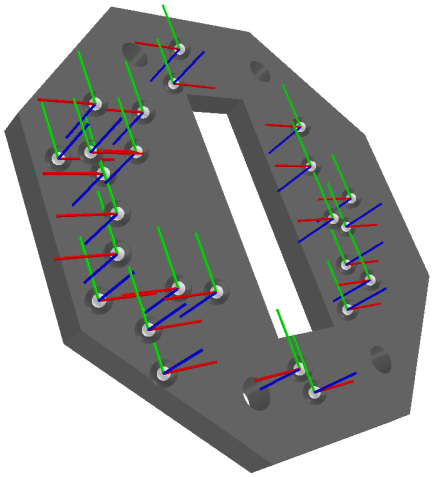


Figure 3.3: IPA Intermediate Plate workpiece

The **IPAIntermediatePlate** as seen in figure 3.3 represents an industrial component with regular geometry and large planar regions. It provides a problem instance representative of structured workpieces commonly found in manufacturing and material-handling applications.

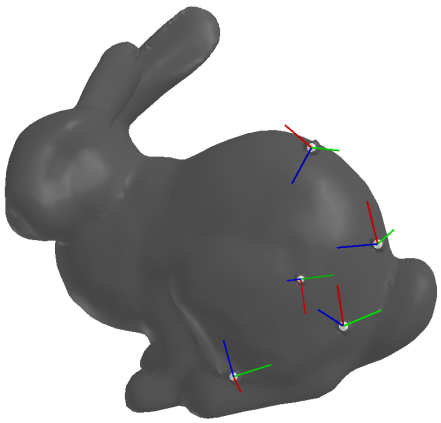


Figure 3.4: Stanford Bunny workpiece

The **Stanford Bunny** as seen in figure 3.4 provides a contrasting geometry with curved

surfaces, irregular features, and greater local geometric variation. It is therefore used to examine algorithm behavior on a less regular workpiece.

Each workpiece defines an independent optimization instance. The workpieces are not combined as objectives in a single optimization run. Their purpose is to evaluate whether the relative behavior of the selected optimization methods changes across different workpiece geometries.

3.6 Evaluation Architecture

Candidate gripper configurations are evaluated through a client–server architecture. The optimization process acts as the client and submits a decision vector $\mathbf{q}$ to the evaluation server. The server evaluates the corresponding gripper geometry and returns the objective vector $\mathbf{f}(\mathbf{q})$.

This separation establishes a strict black-box interface. The optimization algorithms operate only on decision vectors and returned objective values and do not access the internal geometric evaluation procedure. Consequently, algorithms that support the same interface can be exchanged without modifying the geometric evaluation pipelines. The architecture therefore supports modularity, reproducibility, and future extension.

3.7 Comparative Experimental Design

A controlled comparative design is adopted to ensure that observed differences are attributable primarily to the optimization methods. For a given gripper–workpiece instance, all algorithms are evaluated using the same geometric parameterization, objective functions, variable bounds, feasibility conditions, and evaluation budget.

Because the selected optimization methods contain stochastic components, independent repetitions are used to examine performance variability and robustness [94]. Comparisons are performed within the same problem instance. Results from different workpieces or gripper types are not combined into one optimization problem or one Pareto front.

3.8 Performance Indicators

Algorithm performance is evaluated using inverted generational distance plus, the maximum hypervolume attained during a run, and front coverage. These indicators provide complementary information about convergence and the representation of objective-space trade-offs.

IGD+ is selected to assess how well an optimization run approximates the best trade-offs identified for a given gripper–workpiece problem. It is useful in this thesis because it considers both proximity to the reference front and the representation of different regions of that front. This enables the algorithms to be compared based on the overall quality of their approximation sets rather than on individual objective values alone.

Maximum hypervolume is selected as an aggregate measure of approximation-set quality. It provides a single value that reflects both convergence and the extent of the represented objective-space trade-offs. Considering the maximum value attained during a run is particularly relevant for the investigated black-box optimization problems because it identifies the best approximation quality reached within the available optimization budget, even when the quality of the population varies during a run.

Front coverage is selected to determine how broadly an algorithm represents the available trade-off region. This is important for gripper design because a set concentrated in only one region of the Pareto front may provide limited design choice, even if it performs well according to an aggregate indicator. Broad coverage provides a more varied set of candidate designs from which a design can be selected according to application-specific preferences.

The three indicators are complementary. IGD+ measures proximity to and representation of a reference front, maximum hypervolume measures the best combined convergence and diversity reached during a run, and front coverage focuses on the extent of the represented trade-off region. Their values across independent repetitions are used to characterize both optimization performance and its variability.

4 Implementation

4.1 Implementation of Multi-Suction Cup Gripper Optimization

This chapter presents the optimization framework developed for a multi-suction cup gripper consisting of three suction cups arranged in a triangular configuration. The framework supports both evolutionary multi-objective optimization and Bayesian multi-objective optimization. The evolutionary algorithms are implemented using the `pymoo` library [64], whereas Bayesian optimization is implemented using the `Ax` platform and its underlying `BoTorch` modeling framework [3, 91].

For both optimization approaches, the evaluation of candidate gripper layouts is delegated to an external service through an API. This separation provides a modular architecture in which the optimization strategy and the domain-specific grasp evaluation remain decoupled. Consequently, the same decision-variable representation, grasp-generation procedure, and objective functions can be used by both evolutionary and Bayesian optimization methods, allowing their performance to be compared under a common evaluation pipeline.

4.1.1 Reduced Parameterization of Multi-Suction Cup Gripper

The gripper consists of three suction cups whose positions must ultimately be represented in Cartesian space in order to perform grasp generation, collision checking, and objective computation. However, the Cartesian suction cup coordinates are not used directly as optimization variables. Instead, the layout is parameterized in a reduced form to constrain the search space and encode geometric structure.

All three suction cups are constrained to lie on a common circle in the gripper plane. Their configuration is therefore defined by one shared radius r and three angular parameters

α_1,α_2,α_3:

$$\mathbf{q} = [r,\alpha_1,\alpha_2,\alpha_3] \tag{4.1}$$

The angular parameters determine the positions of the suction cups on the circle, while the common radius determines their distance from the gripper center. For a fixed gripper height z_0, the Cartesian coordinates of the suction cups are obtained through the polar-to-Cartesian transformation

$$\mathbf{c}_i = \begin{bmatrix} r\cos(\alpha_i) \\ r\sin(\alpha_i) \\ z_0 \end{bmatrix}, \qquad i = 1,2,3. \tag{4.2}$$

The resulting Cartesian gripper configuration is written as

$$\mathbf{p} = [\mathbf{c}_1,\mathbf{c}_2,\mathbf{c}_3]. \tag{4.3}$$

Thus, the optimization is carried out in the reduced 4-dimensional parameter space $\mathbf{q}$, while the grasp evaluation operates on the derived Cartesian coordinates $\mathbf{p}$.

This formulation reduces the dimensionality of the search space from six planar degrees of freedom,

$$[(x_1,y_1),(x_2,y_2),(x_3,y_3)] \in \mathbb{R}^6, \tag{4.4}$$

to four variables,

$$[r,\alpha_1,\alpha_2,\alpha_3]. \tag{4.5}$$

The reduction improves search efficiency and excludes many geometrically redundant or undesirable layouts while preserving sufficient flexibility for diverse triangular suction cup arrangements.

Assumption A triangular configuration of three suction cups is assumed to yield a stable grasp [87]. This is motivated by the fact that three spatially distributed contact points form a statically stable support polygon, enabling effective load distribution and resistance to external disturbances.

4.1.2 Decision Variables and Bounds for Multi-Suction Cup Gripper

The optimization is performed over the reduced decision vector

$$\mathbf{q} = [r,\alpha_1,\alpha_2,\alpha_3]. \tag{4.6}$$

All variables are treated as continuous. The angular variables are bounded by

$$\alpha_i \in [0,2\pi), \qquad i = 1,2,3, \tag{4.7}$$

and the common radius is bounded by

$$r \in [r_{\min},r_{\max}]. \tag{4.8}$$

The upper bound $r_{\max}$ is determined from the bounding box of the largest workpiece considered during optimization, ensuring that the gripper footprint remains compatible with the object dimensions. The lower bound $r_{\min}$ is determined from the diameter of an individual suction cup, ensuring that the configuration remains physically realizable and that the suction cups do not collapse into an excessively compact arrangement.

All optimization methods generate candidate values within these bounds. In the evolutionary algorithms, the initial population is obtained by random sampling and subsequently modified through the operators of the selected algorithm. In Bayesian optimization, Ax first evaluates the center of the search space and then uses Sobol sampling to generate an initial space-filling design. After sufficient initialization data have been collected, Ax transitions to model-based Bayesian optimization.

4.1.3 Optimization Framework of Multi-Suction Cup Gripper

The multi-suction cup gripper problem is solved using either an evolutionary multi-objective optimization algorithm or Bayesian multi-objective optimization. Both approaches operate on the same reduced decision vector, variable bounds, objective functions, and external evaluation service.

The common optimization configuration contains:

- Number of objectives n_{obj}
- Decision-variable definitions and bounds
- Optimization direction
- Choice of optimization method
- Evaluation endpoint URL

For evolutionary optimization, the additional configuration parameters include:

- Population size N
- Maximum number of generations $G_{\max}$
- Choice of evolutionary algorithm
- Algorithm-specific parameters, such as crossover and mutation settings

For Bayesian optimization, the additional configuration parameters include:

- Number of optimization batches $B_{\max}$
- Maximum number of candidates per batch b
- Bayesian generation strategy

In the current Bayesian implementation, the configuration value named `generations` represents the number of candidate-generation batches rather than evolutionary generations. The maximum requested number of Bayesian evaluations is therefore

$$E_{\max}^{\mathrm{BO}} = B_{\max} b. \tag{4.9}$$

The actual number may be smaller if Ax returns fewer than b candidates in a particular batch.

Supported Optimization Methods

The framework supports the following evolutionary and Bayesian multi-objective optimization methods:

Table 4.1: Supported multi-objective optimization methods.

Family	**Identifier**	**Method**
Evolutionary	NSGA2	Non-dominated Sorting Genetic Algorithm II [61]
Evolutionary	NSGA3	Non-dominated Sorting Genetic Algorithm III [62]
Evolutionary	MOEAD	Multi-Objective Evolutionary Algorithm based on Decomposition [63]
Evolutionary	CMOPSO	Competitive Multi-Objective Particle Swarm Optimization
Evolutionary	SMSEMOA	S-Metric Selection Evolutionary Multi-Objective Algorithm [65]
Bayesian	AX-MBM	Ax modular BoTorch model for multi-objective Bayesian optimization [3, 91]

For evolutionary algorithms requiring reference directions, such as NSGA-III and MOEA/D, uniform reference directions are generated using the method of Das and Dennis [45]:

$$\Lambda = \texttt{get_reference_directions}(\text{“das-dennis”}, n_{\mathrm{obj}}, n_{\mathrm{partitions}}). \tag{4.10}$$

The Bayesian implementation uses the default Ax generation strategy identified at runtime as `Center+Sobol+MBM:fast`. This strategy consists of three stages:

1. Evaluation of the center of the bounded search space

2. Sobol quasi-random sampling for initialization

3. Model-based optimization using the Ax modular BoTorch model (MBM)

The Center and Sobol stages do not employ a surrogate model or acquisition function. After transitioning to the MBM stage, Ax fits probabilistic surrogate models to the completed observations and selects new candidates using an acquisition function appropriate for the multi-objective problem. For continuous multi-objective problems, this generally involves Gaussian-process surrogates and a hypervolume-improvement acquisition function. The exact model and acquisition-function classes are selected by Ax according to the installed version and experiment configuration and should therefore be recorded from the runtime configuration to ensure reproducibility.

4.1.4 Initialization of Multi-Suction Cup Gripper Optimization

The initialization procedure depends on the selected optimization approach.

Evolutionary Initialization

For evolutionary optimization, an initial population of N candidate solutions is generated by sampling the reduced decision variables within their bounds:

$$\mathcal{P}^{(0)} = \{\mathbf{q}_1,\mathbf{q}_2,\ldots,\mathbf{q}_N\}. \tag{4.11}$$

Each candidate decision vector $\mathbf{q}_k$ is transformed into its corresponding Cartesian gripper configuration $\mathbf{p}_k$ before objective evaluation.

Bayesian Initialization

Bayesian optimization does not maintain an evolutionary population. Instead, it stores a dataset of completed trials:

$$\mathcal{D}_t = \{(\mathbf{q}_k,\mathbf{g}(\mathbf{q}_k))\}_{k=1}^{t}\,, \tag{4.12}$$

where $\mathbf{q}_k$ is an evaluated gripper parameterization and $\mathbf{g}(\mathbf{q}_k)$ is its observed objective vector.

The Ax generation strategy first evaluates the center of the search space and then generates an initial set of candidates using Sobol quasi-random sampling. Sobol sampling provides broad coverage of the bounded decision space before a surrogate model is fitted. Once the initialization requirements of the generation strategy have been satisfied, Ax transitions to model-based candidate generation.

4.1.5 Objective Function Evaluation of Multi-Suction Cup Gripper via API

The evaluation of objective values is decoupled from the optimization loop and delegated to an external evaluation service. For each candidate solution $\mathbf{q}$, the corresponding Cartesian suction cup coordinates are first computed by transforming the radial and angular parameters into 3D positions. The resulting Cartesian gripper configuration is then passed to the evaluation endpoint, which executes the grasp evaluation procedure and returns the objective values.

This process can be written as

$$\mathbf{p} = T(\mathbf{q}), \qquad \mathbf{f}(\mathbf{q}) = \mathbf{f}(T(\mathbf{q})) = \texttt{API_call}(\mathbf{p}) \rightarrow [f_1(\mathbf{p}),\, f_2(\mathbf{p})]. \tag{4.13}$$

where $T(\,\cdot\,)$ denotes the transformation from the reduced parameterization to Cartesian suction cup coordinates.

In the implementation, this transformation is carried out programmatically in the evaluation pipeline before grasp generation and objective computation. For a candidate radius and list of angles, the suction cup coordinates are constructed as

$$\mathbf{c}_i = (r\cos(\alpha_i),\, r\sin(\alpha_i),\, z_0), \qquad i = 1,2,3. \tag{4.14}$$

This architecture enables:

- Separation of concerns between optimization logic and domain-specific evaluation
- Scalability through independent deployment of the evaluation service

- Reusability of the optimization framework for different problem domains

Grasp Evaluation

The evaluation service receives a Cartesian gripper configuration $\mathbf{p} = [\mathbf{c}_1,\mathbf{c}_2,\mathbf{c}_3]$ and computes the associated grasp quality measures through the following procedure.

1. **Candidate Grasp Generation and Feasibility Filtering**

   Suction grasp candidates are generated using a method developed by Fraunhofer IPA, following the automatic suction grasp generation procedure described by [68]. Points are sampled on the surface of the workpiece, and each sampled point defines a candidate grasp through its position and local surface normal. The local surface normal determines the approach direction of the suction cup.

   In the original method, each suction grasp candidate is tested for feasible seal formation using a quasi-static spring model of a suction cup, as introduced in [69]. If the grasp point is considered feasible, the candidate is further tested with a gripper collider model for collisions between the gripper and the object.

   In this thesis, the method was extended to a gripper with three suction cups. For each candidate gripper pose, the seal-formation test is performed for all three suction cup contact positions $\mathbf{c}_1$, $\mathbf{c}_2$, and $\mathbf{c}_3$. A candidate is considered feasible only if all three suction cups form a sufficient seal. The candidate is then checked for collisions in simulation, and only collision-free candidates are retained as valid grasps.

2. **Clustering of Grasps**

   The set $\mathcal{G}_{\text{valid}}$ often contains many grasps that are nearly identical in position and orientation. When the number of valid grasps $m = |\mathcal{G}_{\text{valid}}|$ exceeds a threshold `nb_grasps_clustering`, a farthest-point clustering procedure is applied to select a diverse reduced subset. Otherwise, all valid grasps are used directly.

   Each grasp g_i is represented by a pose descriptor

   $$\mathbf{z}_i = [\,\mathrm{vec}(\mathbf{R}_i)^\top,\, \mathbf{t}_i^\top\,]^\top \in \mathbb{R}^{12}, \tag{4.15}$$

where $\mathbf{R}_i \in SO(3)$ is the orientation and $\mathbf{t}_i \in \mathbb{R}^3$ is the position. Two distances are used: the translational distance

$$d_{\text{pos}}(i,j) = \|\mathbf{t}_i - \mathbf{t}_j\|_2 \tag{4.16}$$

and the orientation distance

$$d_{\text{ori}}(i,j) = \frac{1}{9}\|\text{vec}(\mathbf{R}_i) - \text{vec}(\mathbf{R}_j)\|_2^2. \tag{4.17}$$

Cluster centers are selected greedily by farthest-point sampling in position space. The first center is the grasp with the largest mean translational distance to all others, and each subsequent center maximizes its minimum distance to the already selected centers. This promotes broad spatial coverage.

Optionally, orientation diversity is added within each center's local neighborhood. For a selected center m_k, grasps within a spatial radius δ_{intra} form a neighborhood, from which up to q_{intra} additional grasps are chosen by farthest-point sampling on d_{ori}, capturing distinct orientations at similar positions.

The resulting clustered subset

$$\mathcal{G}_{\text{clustered}} \subseteq \mathcal{G}_{\text{valid}} \tag{4.18}$$

is then used for diversity evaluation.

3. **Triangle Area Calculation**

The area of the triangle formed by the three suction cups is computed from their Cartesian coordinates. If the planar coordinates of the three cups are (x_1,y_1), (x_2,y_2), and (x_3,y_3), then

$$A(\mathbf{p}) = \frac{1}{2}\left|(x_2 - x_1)(y_3 - y_1) - (x_3 - x_1)(y_2 - y_1)\right|. \tag{4.19}$$

This objective favors gripper layouts with a wider support polygon, which is associated with improved grasp stability.

4. **Diversity Metric Calculation**

A diversity score $D(\mathbf{p})$ is computed to quantify how broadly the resulting valid grasp set is distributed in position and orientation.

Let the set used for diversity evaluation contain n grasps. If clustering is performed, then

$$n = |\mathcal{G}_{\text{clustered}}|, \tag{4.20}$$

otherwise

$$n = |\mathcal{G}_{\text{valid}}|. \tag{4.21}$$

Let the corresponding grasp positions be

$$\{\mathbf{y}_1, \ldots, \mathbf{y}_n\} \tag{4.22}$$

and the corresponding orientation matrices be

$$\{\mathbf{R}_1, \ldots, \mathbf{R}_n\}. \tag{4.23}$$

To normalize the spatial term with respect to object size, a characteristic object scale is defined as the Euclidean norm of the axis-aligned bounding-box extents:

$$s = \|\mathbf{e}\|_2, \tag{4.24}$$

where $\mathbf{e} \in \mathbb{R}^3$ contains the bounding-box side lengths.

The spatial component of diversity is computed as the mean pairwise Euclidean distance between grasp positions:

$$D_{\text{pos}}(\mathbf{p}) = \frac{1}{s} \cdot \frac{1}{n(n-1)} \sum_{\substack{i,j=1 \\ i \neq j}}^{n} \|\mathbf{y}_i - \mathbf{y}_j\|_2. \tag{4.25}$$

The orientation component is computed from the mean pairwise geodesic distance between rotation matrices. For two grasp orientations $\mathbf{R}_a, \mathbf{R}_b \in SO(3)$, the relative

rotation is

$$\Delta\mathbf{R} = \mathbf{R}_a^\top \mathbf{R}_b \tag{4.26}$$

and the associated geodesic distance is

$$d_{\text{geo}}(\mathbf{R}_a,\mathbf{R}_b) = \cos^{-1}\left(\frac{\text{tr}(\Delta\mathbf{R})-1}{2}\right). \tag{4.27}$$

For numerical stability, the argument of $\cos^{-1}$ is clipped to the interval $[-1,1]$.

The normalized orientation diversity is then defined as

$$D_{\text{rot}}(\mathbf{p}) = \frac{1}{\pi} \cdot \frac{2}{n(n-1)} \sum_{1\le i<j\le n} d_{\text{geo}}(\mathbf{R}_i,\mathbf{R}_j). \tag{4.28}$$

The final diversity score is computed as the equally weighted combination

$$D(\mathbf{p}) = \text{clip}\left(0.5\, D_{\text{pos}}(\mathbf{p}) + 0.5\, D_{\text{rot}}(\mathbf{p}), 0, 1\right). \tag{4.29}$$

Continuity considerations Clustering is only applied when the number of valid grasps exceeds the predefined threshold `nb_grasps_clustering`. If fewer valid grasps are available, clustering is skipped and the diversity metric is evaluated directly on the valid grasp set. Therefore, the omission of clustering does not by itself introduce a discontinuity in the objective function.

A discontinuity may arise only in the degenerate case where fewer than two valid grasps remain after collision filtering. Since both the spatial and orientation components of the diversity metric are based on pairwise distances, these terms cannot be evaluated meaningfully for $n < 2$. In that case, the corresponding terms are set to zero. As a result, small changes in the gripper configuration may cause an abrupt transition between a nonzero diversity score and a degenerate zero-valued diversity contribution.

To reduce the likelihood of this case, the number of initially sampled surface points was increased from 100 to 1000. The denser surface sampling produces a larger set of

candidate grasps and makes it much less likely that fewer than two valid grasps remain after filtering. Consequently, the diversity metric is evaluated on a sufficiently rich grasp set in the vast majority of cases, yielding a more stable optimization objective in practice.

Objective Evaluation

After grasp evaluation, each candidate layout is assigned the natural objective vector

$$\mathbf{g}(\mathbf{q}) = [A(\mathbf{p}), D(\mathbf{p})], \qquad \mathbf{p} = T(\mathbf{q}). \tag{4.30}$$

Both components are maximized. A larger triangle area is associated with a wider support polygon and improved grasp stability, while a larger diversity score indicates that the valid grasps are more broadly distributed in position and orientation.

The objective-direction convention is handled differently by the two optimization frameworks. Since `pymoo` assumes minimization, the evolutionary algorithms negates objective vector

$$\mathbf{f}_{\mathrm{EA}}(\mathbf{q}) = [-A(\mathbf{p}),\ -D(\mathbf{p})]. \tag{4.31}$$

In the Bayesian implementation, the external evaluation service values are left as is, since `Ax` assumes maximization

$$\mathbf{f}_{\mathrm{BO}}(\mathbf{q}) = [A(\mathbf{p}), D(\mathbf{p})]. \tag{4.32}$$

The optimization direction is specified through the Ax objective expression. Consequently, the observations themselves are not negated before being reported to Ax. This preserves the original physical meaning of the objective values in the returned Pareto set.

4.1.6 Optimization Loop for Multi-Suction Cup Gripper

The candidate-generation process differs between evolutionary and Bayesian optimization, while the external candidate-evaluation procedure remains identical.

Evolutionary Optimization Loop

The evolutionary optimization proceeds according to

$$\mathcal{P}^{(t+1)} = \mathcal{A}\left(\mathcal{P}^{(t)}, \mathbf{f}_{\mathrm{EA}}\right), \qquad t = 0,1,\ldots, \tag{4.33}$$

where $\mathcal{A}$ denotes the selected evolutionary algorithm.

Depending on the selected algorithm, the new population is generated through operations such as selection, crossover, mutation, decomposition, particle movement, or survival based on hypervolume contribution. Every candidate in the resulting population is evaluated through the common API-based grasp evaluation pipeline.

Bayesian Optimization Loop

The Bayesian optimization procedure maintains the observation dataset $\mathcal{D}_t$. During the initialization phase, candidates are generated from the center of the search space and by Sobol sampling. During the model-based phase, the available observations are used to fit probabilistic surrogate models:

$$\widehat{\mathbf{g}}_t = \mathcal{M}(\mathcal{D}_t), \tag{4.34}$$

where $\mathcal{M}$ denotes the surrogate-modeling procedure used by Ax and BoTorch.

The surrogate models provide predictions of the objective values together with their uncertainty. An acquisition function $\alpha(\mathbf{q} \mid \mathcal{D}_t)$ uses this information to balance the exploration of uncertain regions and the exploitation of regions expected to improve the current Pareto frontier.

For a requested batch size b, the next candidate batch is generated as

$$\mathcal{Q}_{t+1} = \underset{\substack{\mathcal{Q}\subset\mathcal{X}\\|\mathcal{Q}|\leq b}}{\arg\max}\, \alpha(\mathcal{Q} \mid \mathcal{D}_t), \tag{4.35}$$

where $\mathcal{X}$ is the bounded decision space.

Each candidate $\mathbf{q} \in \mathcal{Q}_{t+1}$ is evaluated through the external evaluation service. The resulting observation is then supplied to Ax as a completed trial:

$$\mathcal{D}_{t+1} = \mathcal{D}_t \cup \{(\mathbf{q}_{t+1}, \mathbf{g}(\mathbf{q}_{t+1}))\}\,. \tag{4.36}$$

The surrogate model is subsequently updated using the expanded dataset, and the process is repeated until the specified evaluation budget is exhausted. Although multiple candidates can be requested in one batch, the current implementation submits the candidates in that batch sequentially to the evaluation service.

4.1.7 Termination Criteria for Multi-Suction Cup Gripper Optimization

The termination criteria depend on the selected optimization approach.

Table 4.2: Termination criteria for evolutionary and Bayesian optimization.

Approach	**Criterion**	**Symbol**	**Description**
Evolutionary	Design-space tolerance	x_{tol}	Change in decision variables below threshold
Evolutionary	Objective-space tolerance	f_{tol}	Relative objective improvement below threshold
Evolutionary	Maximum generations	$G_{\max}$	Maximum number of generations reached
Evolutionary	Maximum evaluations	$E_{\max}$	Maximum number of function evaluations reached
Evolutionary	Hypervolume	$HV_{\max}$	Hypervolume progression stagnates
Bayesian	Maximum batches	$B_{\max}$	Maximum number of candidate-generation batches reached
Bayesian	Maximum evaluations	$E^{\text{BO}}_{\max}$	Bayesian evaluation budget exhausted

For evolutionary optimization, the design-space and objective-space tolerances are evaluated over a sliding window of w generations.

In the current Bayesian implementation, termination is budget-based. The optimization performs at most $B_{\max}$ candidate-generation batches, with up to b candidates requested per batch. Therefore,

$$E_{\max}^{\mathrm{BO}} = B_{\max} b. \tag{4.37}$$

No design-space or objective-space convergence tolerance is currently applied to the Bayesian optimization loop.

4.1.8 Performance Monitoring for Multi-Suction Cup Gripper

The hypervolume indicator [95] is tracked at each generation via a callback mechanism:

$$\mathrm{HV}^{(t)} = \mathrm{HV}(\mathcal{F}^{(t)}, \mathbf{r}) \tag{4.38}$$

Here, $\mathbf{r} \in \mathbb{R}^{n_{\mathrm{obj}}}$ is the reference point and $\mathcal{F}^{(t)}$ denotes the set of objective vectors at generation t. A monotonically increasing hypervolume indicates convergence toward the Pareto front.

4.1.9 Optional Hyperparameter Optimization for Multi-Suction Cup Gripper Problem

The Evolutionary Optimization framework optionally supports automated tuning of algorithm hyperparameters, such as crossover probability and mutation rate, using a meta-optimization approach. A `HyperparameterProblem` is constructed around the selected algorithm and evaluated over multiple random seeds. The mean hypervolume across runs serves as the meta-objective:

$$\min_{\boldsymbol{\theta}} \left\{ -\frac{1}{|\mathcal{Z}|} \sum_{s \in \mathcal{Z}} \mathrm{HV}(\mathcal{F}_s, \mathbf{r}) \right\} \tag{4.39}$$

The best hyperparameters are then applied to the main optimization run.

4.1.10 Output of Multi-Suction Cup Gripper Optimization

The optimization produces a set of Pareto-optimal solutions:

$$\mathcal{P}^* = \{\mathbf{q} \mid \nexists\, \mathbf{q}' \in \mathcal{P} : \mathbf{f}(\mathbf{q}') \preceq \mathbf{f}(\mathbf{q})\}\,. \tag{4.40}$$

The final output includes:

- The Pareto-optimal set $\mathcal{P}^*$ in the reduced decision space, with corresponding objective values
- The associated radial gripper parameterizations $\mathbf{q} = [r,\alpha_1,\alpha_2,\alpha_3]$
- The hypervolume progression $\{\mathrm{HV}^{(0)},\mathrm{HV}^{(1)},\ldots,\mathrm{HV}^{(T)}\}$
- Total optimization wall-clock time

These solutions represent optimal trade-offs between grasp diversity and triangle area, i.e., grasp robustness and stability.

4.2 Implementation of Two-Finger Gripper Optimization

This chapter presents the optimization framework developed for a two-finger gripper with a spline-based gripping contour. The framework supports both evolutionary multi-objective optimization and Bayesian multi-objective optimization. The evolutionary algorithms are implemented using the `pymoo` library [64], while Bayesian optimization is implemented using the `Ax` platform and its underlying `BoTorch` modeling framework [3, 91].

For both approaches, the evaluation of candidate gripper designs is delegated to an external service through an API. The evaluation service constructs the gripper geometry, voxelizes the gripper and workpieces, and calculates the grasp-quality objectives using a modified

version of Snuggle-Pack[90]. This separation provides a modular architecture in which the optimization logic and the domain-specific grasp evaluation remain decoupled.

The same decision variables, parameter bounds, geometry-generation pipeline, workpieces, orientations, and grasp-quality objectives are used for both optimization approaches. This enables their performance to be compared under a common evaluation budget and evaluation procedure.

4.2.1 Decision Variables and Bounds for Two-Finger Gripper Problem

A candidate gripper is represented by the decision vector

$$\mathbf{x} = \begin{bmatrix} d_{\text{gn}} & l_{\text{gn}} & h_1 & h_2 & h_3 \end{bmatrix}^{\text{T}}, \tag{4.41}$$

where the variables are transmitted to the evaluation API in the order

$$\mathbf{x} = [\texttt{depth_gn}, \texttt{length_gn}, \texttt{h_p1_spline}, \texttt{h_p2_spline}, \texttt{h_p3_spline}]^{\text{T}}. \tag{4.42}$$

The decision variables and their bounds are summarized in Table 4.3.

Table 4.3: Decision variables of the two-finger gripper optimization problem.

Variable	**Configuration parameter**	**Lower bound**	**Upper bound**
d_{gn}	`depth_gn`	1	25
l_{gn}	`length_gn`	1	50
h_1	`h_p1_spline`	1	15
h_2	`h_p2_spline`	1	15
h_3	`h_p3_spline`	1	15

All five decision variables are treated as continuous variables. Candidate values generated by either optimization framework are restricted to the bounded search space

$$\mathcal{X} = [1{,}25] \times [1{,}50] \times [1{,}15]^3. \tag{4.43}$$

For evolutionary optimization, the variables are initially sampled within these bounds and subsequently modified by the operators of the selected algorithm. For Bayesian optimization, Ax initially evaluates the center of the search space and generates additional initialization candidates using Sobol sampling. After sufficient observations have been collected, Ax transitions to model-based candidate generation.

4.2.2 Optimization Framework for Two-Finger Gripper Optimization

The two-finger gripper problem is solved using either an evolutionary multi-objective optimization algorithm or Bayesian multi-objective optimization. Both methods use the same bounded decision space, objective definitions, and external evaluation service.

The common optimization configuration contains:

- Number of objectives n_{obj}
- Decision-variable definitions and bounds
- Optimization direction
- Choice of optimization method
- Evaluation endpoint URL

For evolutionary optimization, the additional configuration parameters include:

- Population size N
- Maximum number of generations $G_{\max}$
- Choice of evolutionary algorithm

For Bayesian optimization, the additional configuration parameters include:

- Maximum number of candidate-generation batches $B_{\max}$
- Maximum number of candidates requested per batch b
- Bayesian generation strategy

In the Bayesian implementation, a batch represents a candidate-generation step rather than an evolutionary generation. The maximum requested number of Bayesian evaluations is therefore

$$E_{\max}^{\mathrm{BO}} = B_{\max} b. \tag{4.44}$$

The actual number of evaluations may be smaller if the generation strategy returns fewer than b candidates in a batch.

Supported Optimization Methods for Two-Finger Gripper Optimization

The framework supports the evolutionary and Bayesian multi-objective optimization methods listed in Table 4.4.

Table 4.4: Supported multi-objective optimization methods for the two-finger gripper.

Family	**Identifier**	**Method**
Evolutionary	NSGA2	Non-dominated Sorting Genetic Algorithm II [61]
Evolutionary	NSGA3	Non-dominated Sorting Genetic Algorithm III [62]
Evolutionary	MOEAD	Multi-Objective Evolutionary Algorithm based on Decomposition [63]
Evolutionary	CMOPSO	Competitive Multi-Objective Particle Swarm Optimization
Evolutionary	SMSEMOA	S-Metric Selection Evolutionary Multi-Objective Algorithm [65]
Bayesian	AX-MBM	Ax modular BoTorch model for multi-objective Bayesian optimization [3, 91]

For evolutionary algorithms requiring reference directions, such as NSGA-III and MOEA/D, uniform reference directions are generated using the method of Das and Dennis [45]:

$$\Lambda = \texttt{get_reference_directions}(\text{“das-dennis”}, n_{\mathrm{obj}}, n_{\mathrm{partitions}}). \tag{4.45}$$

The Bayesian implementation uses the Ax generation strategy identified at runtime as `Center+Sobol+MBM:fast`. This strategy consists of the following stages:

1. Evaluation of the center of the bounded decision space

2. Sobol quasi-random sampling for initial space exploration

3. Model-based candidate generation using the Ax modular BoTorch model (MBM)

The Center and Sobol stages do not use a surrogate model or an acquisition function. Once the generation strategy transitions to the MBM stage, Ax uses the available observations to fit probabilistic surrogate models and optimize an acquisition function for multi-objective candidate selection. For continuous multi-objective problems, this commonly involves Gaussian-process surrogates and a hypervolume-improvement acquisition function such as `qLogNEHVI`.

The exact surrogate and acquisition-function classes are selected by Ax and can depend on the installed Ax and BoTorch versions, the number of observations, and the experiment configuration. Therefore, the active generation stage, surrogate model, and acquisition function should be recorded from the runtime configuration for reproducibility.

4.2.3 Initialization of Two-Finger Gripper Optimization

The initialization process depends on whether evolutionary or Bayesian optimization is selected.

Evolutionary Initialization

The selected evolutionary algorithm initializes a population of N candidate grippers:

$$\mathcal{P}^{(0)} = \left\{ \mathbf{x}_1^{(0)}, \mathbf{x}_2^{(0)}, \ldots, \mathbf{x}_N^{(0)} \right\}, \qquad \mathbf{x}_i^{(0)} \in \mathcal{X}. \tag{4.46}$$

Each candidate is sampled within the variable bounds. The complete population is evaluated through the external API before the selected evolutionary algorithm performs its first selection and variation step.

Bayesian Initialization

Bayesian optimization does not maintain an evolutionary population. Instead, it maintains a dataset containing the completed gripper evaluations:

$$\mathcal{D}_t = \{(\mathbf{x}_i, \mathbf{k}(\mathbf{x}_i))\}_{i=1}^{t}, \tag{4.47}$$

where

$$\mathbf{k}(\mathbf{x}) = \begin{bmatrix} k_1(\mathbf{x}) & k_2(\mathbf{x}) & \cdots & k_{n_{\text{obj}}}(\mathbf{x}) \end{bmatrix}^{\mathrm{T}} \tag{4.48}$$

contains the grasp-quality KPIs returned by the evaluation service.

The Ax generation strategy initially evaluates the center of the bounded search space. Additional initialization candidates are generated using Sobol quasi-random sampling. Sobol sampling provides broad and approximately space-filling coverage of the five-dimensional search space before a surrogate model is introduced.

After the initialization requirements of the generation strategy have been satisfied, Ax transitions to the model-based MBM stage. From this point onward, new gripper designs are selected using surrogate predictions and predictive uncertainty.

4.2.4 Objective Function Evaluation of Two-Finger Gripper via API

For every candidate, the decision variables are sent to the evaluation API as an ordered list. The API inserts these values into a complete gripper configuration:

$$\mathbf{x} \longrightarrow \mathbf{c}_{\text{gripper}}(\mathbf{x}) \longrightarrow \mathcal{G}(\mathbf{x}), \tag{4.49}$$

where $\mathbf{c}_{\text{gripper}}$ denotes the gripper configuration and $\mathcal{G}(\mathbf{x})$ the resulting geometry.

The configuration also contains fixed parameters, including the contour type, basic jaw dimensions, jaw angle, stroke, and tool-center-point translation. The spline contour is selected by the static configuration and is not itself an optimization variable. The completed configuration is used to create the jaw geometry using CadQuery [96] which converts it into the mesh representation required for evaluation.

Grasp Evaluation

The generated gripper is evaluated against each configured workpiece. The gripper and workpiece geometries are then voxelized to obtain three-dimensional occupancy grids.

The grasp evaluation uses a modified version of Snuggle-Pack, a three-dimensional packing algorithm that combines FFT-based spatial analysis with proximity heuristics [90]. In the original method, correlations between voxel grids are used to identify collision-free object placements. Here, the same principle is adapted to evaluate relative gripper–workpiece placements.

Let $s_{\mathcal{G}}$ and $s_{\mathcal{W}}$ denote the occupancy grids of the gripper and workpiece. Their spatial correlation produces a collision map:

$$\zeta_{\mathcal{G},\mathcal{W}}(\mathbf{q}) = \mathcal{F}^{-1}\left(\overline{\mathcal{F}(s_{\mathcal{G}})}\,\mathcal{F}(s_{\mathcal{W}})\right)(\mathbf{q}), \tag{4.50}$$

where $\mathbf{q}$ is a relative displacement and $\mathcal{F}$ denotes the Fourier transform. Values greater than zero indicate overlapping voxels, whereas zero values identify collision-free relative placements.

The collision map is combined with Gaussian proximity distributions to determine how closely the fingers surround the workpiece at feasible placements. These calculations are performed using FFT-based convolution and GPU-accelerated PyTorch operations. The resulting grasp quality is represented by the KPI `max_prox`. If the required opening cannot be achieved within the available gripper stroke, the KPI is set to zero.

Objective Evaluation

The evaluation service returns one grasp-quality KPI for each evaluated workpiece–orientation combination. Since a larger proximity KPI represents a better grasp result, the natural optimization problem is

$$\max_{\mathbf{x}\in\mathcal{X}} k_m(\mathbf{x}), \qquad m = 1,\ldots,n_{\text{obj}}. \tag{4.51}$$

The natural objective vector is therefore

$$\mathbf{k}(\mathbf{x}) = \begin{bmatrix} k_1(\mathbf{x}) & k_2(\mathbf{x}) & \cdots & k_{n_{\text{obj}}}(\mathbf{x}) \end{bmatrix}^{\mathrm{T}}. \tag{4.52}$$

The objective-direction convention is handled differently by the two optimization frameworks. Since `pymoo` assumes minimization, the evolutionary algorithms negates the objective vector

$$\mathbf{f}_{\text{EA}}(\mathbf{x}) = -\mathbf{k}(\mathbf{x}) = \begin{bmatrix} -k_1(\mathbf{x}) & -k_2(\mathbf{x}) & \cdots & -k_{n_{\text{obj}}}(\mathbf{x}) \end{bmatrix}^{\mathrm{T}}. \tag{4.53}$$

In the Bayesian implementation, the external evaluation service values are left as is, since `Ax` assumes maximization:

$$\mathbf{f}_{\text{BO}}(\mathbf{x}) = \mathbf{k}(\mathbf{x}). \tag{4.54}$$

The maximization direction is specified in the Ax optimization configuration. Consequently, the observed KPI values do not need to be negated before they are attached to the corresponding Ax trial. This also preserves the physical meaning of the KPI values when the final Pareto frontier is returned.

If n_W workpieces and n_R orientations are evaluated separately, the number of objectives is

$$n_{\text{obj}} = n_W n_R. \tag{4.55}$$

For the currently configured single orientation, each workpiece contributes one objective:

$$n_{\text{obj}} = n_W. \tag{4.56}$$

For performance comparisons, the results from both optimization frameworks are converted to the common natural maximization convention $\mathbf{k}(\mathbf{x})$.

4.2.5 Optimization Loop for Two-Finger Gripper

The external evaluation pipeline is shared by the evolutionary and Bayesian optimization approaches. The methods differ primarily in how they use the completed evaluations to generate subsequent candidates.

For every candidate, the common evaluation steps are:

1. Send the candidate decision vector to the external evaluation API.
2. Map the decision variables to the complete gripper configuration.
3. Generate the spline-based gripper geometry using CadQuery.
4. Voxelize the gripper and configured workpieces.
5. Calculate FFT-based collision maps using the modified Snuggle-Pack procedure.
6. Calculate one proximity KPI for each workpiece–orientation combination.
7. Return the KPI vector to the selected optimization framework.

Evolutionary Optimization Loop

In every evolutionary generation, the selected algorithm generates a population of candidate decision vectors. The candidates are evaluated through the common evaluation pipeline, and the resulting KPI values are negated before being supplied to `pymoo`.

The population update can be expressed abstractly as

$$\mathcal{P}^{(g+1)} = \mathcal{A}\left(\mathcal{P}^{(g)}, \mathbf{F}_{\mathrm{EA}}^{(g)}\right), \tag{4.57}$$

where $\mathcal{A}$ represents the selected evolutionary algorithm and $\mathbf{F}_{\mathrm{EA}}^{(g)}$ contains the minimization-form objective vectors evaluated in generation g.

Depending on the selected algorithm, the next population is produced using operations such as selection, crossover, mutation, decomposition, particle movement, or survival based on hypervolume contribution.

Bayesian Optimization Loop

During the model-based stage, Ax and BoTorch use the completed trial dataset $\mathcal{D}_t$ to fit probabilistic surrogate models:

$$\widehat{\mathbf{k}}_t = \mathcal{M}(\mathcal{D}_t), \tag{4.58}$$

where $\mathcal{M}$ denotes the surrogate-modeling procedure.

The surrogate models approximate the relationship between the gripper parameters and the workpiece-specific KPI values. In addition to predicted objective values, they estimate predictive uncertainty in regions containing few or no observations.

An acquisition function $\alpha(\mathbf{x} \mid \mathcal{D}_t)$ uses these predictions to balance:

- Exploitation of gripper designs expected to improve the current Pareto frontier
- Exploration of uncertain regions of the bounded decision space

For a requested batch size b, the next candidate batch is generated as

$$\mathcal{X}_{t+1} = \underset{\substack{\mathcal{B}\subset\mathcal{X} \\ |\mathcal{B}|\leq b}}{\arg\max}\, \alpha(\mathcal{B} \mid \mathcal{D}_t). \tag{4.59}$$

Each generated candidate is evaluated using the external service. The KPI observation is then attached to the corresponding Ax trial:

$$\mathcal{D}_{t+1} = \mathcal{D}_t \cup \{(\mathbf{x}_{t+1}, \mathbf{k}(\mathbf{x}_{t+1}))\}\,. \tag{4.60}$$

After new trials have been completed, the surrogate models are updated using the expanded dataset, and another candidate batch is generated. This process continues until the Bayesian evaluation budget is exhausted.

Although multiple candidates can be requested in one Bayesian batch, the current implementation evaluates the candidates one after another through the external API. The implementation therefore performs *sequential batch evaluation*, rather than parallel batch evaluation.

4.2.6 Termination Criteria for Two-Finger Gripper Optimization

The termination criteria depend on the selected optimization approach.

Table 4.5: Termination criteria for evolutionary and Bayesian optimization.

Approach	**Criterion**	**Symbol**	**Description**
Evolutionary	Design-space tolerance	x_{tol}	Change in decision variables below threshold
Evolutionary	Objective-space tolerance	f_{tol}	Relative objective improvement below threshold
Evolutionary	Maximum generations	$G_{\max}$	Maximum number of generations reached
Evolutionary	Maximum evaluations	$E_{\max}$	Maximum number of evaluations reached
Evolutionary	Hypervolume	$HV_{\max}$	Hypervolume progression stagnates
Bayesian	Maximum batches	$B_{\max}$	Maximum number of candidate batches reached
Bayesian	Maximum evaluations	$E_{\max}^{\text{BO}}$	Bayesian evaluation budget exhausted

For evolutionary optimization, the design-space and objective-space tolerances are evaluated over a sliding window of w generations.

The current Bayesian implementation uses budget-based termination. It performs at most $B_{\max}$ candidate-generation batches and requests up to b candidates in each batch. Therefore,

$$E_{\max}^{\text{BO}} = B_{\max} b. \tag{4.61}$$

4.2.7 Performance Monitoring for Two-Finger Gripper

The hypervolume indicator [95] is used to monitor both evolutionary and Bayesian optimization. Before calculating hypervolume, all objective vectors are represented using the same natural maximization convention:

$$\mathbf{k}(\mathbf{x}) = [k_1(\mathbf{x}), \ldots, k_{n_{\text{obj}}}(\mathbf{x})]^{\text{T}}. \tag{4.62}$$

For evolutionary optimization, a callback records the optimization state after each generation. Let $\mathcal{K}_{\mathrm{EA}}^{(g)}$ denote the non-dominated KPI vectors available after generation g. The corresponding hypervolume is

$$\mathrm{HV}_{\mathrm{EA}}^{(g)} = \mathrm{HV}\left(\mathcal{K}_{\mathrm{EA}}^{(g)}, \mathbf{r}\right). \tag{4.63}$$

For Bayesian optimization, hypervolume can be calculated after every completed trial or after every completed candidate batch. Let $\mathcal{K}_{\mathrm{BO}}^{(t)}$ denote the non-dominated KPI vectors obtained from the first t completed Bayesian trials. The corresponding hypervolume is

$$\mathrm{HV}_{\mathrm{BO}}^{(t)} = \mathrm{HV}\left(\mathcal{K}_{\mathrm{BO}}^{(t)}, \mathbf{r}\right). \tag{4.64}$$

Under the maximization convention, the fixed reference point $\mathbf{r}$ must be worse than the objective values of interest. The same reference point and objective scaling must be used for all compared methods. An increasing hypervolume generally indicates improvement in convergence toward the Pareto front, coverage of the objective space, or both.

Evolutionary generations and Bayesian batches can contain different numbers of candidate evaluations. Therefore, the primary comparison is performed using hypervolume as a function of the cumulative number of objective evaluations:

$$\mathrm{HV} = \mathrm{HV}(E), \tag{4.65}$$

where E is the number of completed external API evaluations.

The framework additionally records total wall-clock time because every function evaluation includes API communication, CadQuery geometry generation, voxelization, and GPU-based FFT spatial analysis. Hypervolume as a function of wall-clock time may therefore also be used to compare practical optimization efficiency.

4.2.8 Optional Hyperparameter Optimization for Two-Finger Gripper Problem

For the evolutionary algorithms, the framework optionally supports automated tuning of algorithm hyperparameters, such as crossover probability and mutation rate, using a meta-

optimization approach. A `HyperparameterProblem` is constructed around the selected algorithm and evaluated over multiple random seeds. The negative mean hypervolume across the runs is used as the meta-objective:

$$\min_{\boldsymbol{\theta}} \left\{ -\frac{1}{|\mathcal{Z}|} \sum_{s \in \mathcal{Z}} \mathrm{HV}(\mathcal{K}_s, \mathbf{r}) \right\}. \tag{4.66}$$

The best hyperparameter configuration is subsequently applied to the main evolutionary optimization run.

The current Bayesian implementation uses the generation strategy, surrogate configuration, and acquisition-function configuration selected by Ax. A separate meta-optimization of Bayesian model hyperparameters is therefore not currently performed. The user specifies the candidate batch size and optimization budget, while Ax manages the transition from initialization to model-based optimization.

For reproducibility, each Bayesian experiment records:

- Ax and BoTorch software versions
- Random seed, where supported
- Generation strategy
- Number of initialization trials
- Candidate batch size
- Total evaluation budget
- Active surrogate and acquisition-function classes after entering MBM

4.2.9 Output of Two-Finger Gripper Optimization

The optimization produces a set of non-dominated gripper configurations. The Pareto-optimal set is defined as

$$\mathcal{P}^* = \left\{ \mathbf{x} \in \mathcal{P} \mid \nexists\, \mathbf{x}' \in \mathcal{P} : \mathbf{x}' \prec \mathbf{x} \right\}, \tag{4.67}$$

where $\mathbf{x}' \prec \mathbf{x}$ means that

$$f_m(\mathbf{x}') \leq f_m(\mathbf{x}) \quad \forall m \tag{4.68}$$

and

$$f_j(\mathbf{x}') < f_j(\mathbf{x}) \quad \text{for at least one objective } j. \tag{4.69}$$

The final output includes:

- The Pareto-optimal decision vectors $\mathbf{x} = [d_{\mathrm{gn}}, l_{\mathrm{gn}}, h_1, h_2, h_3]^{\mathrm{T}}$
- The corresponding objective KPI values
- The hypervolume progression $\{\mathrm{HV}^{(0)}, \mathrm{HV}^{(1)}, \ldots, \mathrm{HV}^{(T)}\}$
- Total optimization wall-clock time

The non-dominated solutions represent the best trade-offs between grasp quality for the configured workpieces and orientations.

5 Validation

5.1 Validation of Results for Suction-Cup Gripper

5.2 Validation of Results for Two-Finger Gripper

6 Evaluation

6.1 Results for Suction-Cup Gripper

6.1.1 Results for Evolutionary Algorithms

6.1.2 Results for Bayesian Algorithms

6.2 Results for Two-Finger Gripper

6.2.1 Results for Evolutionary Algorithms

6.2.2 Results for Bayesian Algorithms

7 Summary

7.1 Conclustion

7.2 Limitations

7.3 Future Work

A Appendix

Alles was den Hauptteil unnï¿½tig vergrï¿½ï¿½ert hï¿½tte, z. B. HW-/SW-Dokumentationen, Bedienungsanleitungen, Code-Listings, Diagramme

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